



Kinarm End-Point Lab (PN 14332)

Hardware Guide





When installed in accordance with this manual and verified by a BKIN technician, the equipment described in this guide is in conformity with the relevant Essential Health and Safety Requirements of the following Directives:

- 2004/108/EC (relating to electromagnetic compatibility)
- 2006/95/EC (relating to electrical equipment designed for use with certain voltage limits)
- 2006/42/EC (on machinery)

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1 Revision History

Revision	Description	Date	Approved By
1	Initial Release	26-Mar-2021	IEB
2	Updates for new revision of Kinarm EP Lab 14332 that contains 15050 Power Console and AV Controller.	28-Oct-2025	IEB

2 Read This First

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Form: BT053 Version: 6-Sep-19

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2.2 Product Disclaimer

Kinarm Standard Tests are currently in research and development and are currently only available to researchers with an IRB/REB approved protocol.

****Intended Use**

Kinarm Standard Tests are intended as research tools to contribute to the understanding of brain function and dysfunction. Kinarm Standard Tests are intended to be used on a Kinarm Exoskeleton Lab and/or a Kinarm End-Point Lab. Kinarm Standard Tests do not directly offer a medical diagnosis of any type, nor are Kinarm Standard Tests to be used as an assessment tool to assist with diagnosis. A diagnosis of any brain injury or disease can be made only by a qualified physician or psychologist.

****Contraindications**

Kinarm Standard Tests are not indicated for research subjects who do not have adequate cognitive functioning to understand the task instructions, who do not have visual and/or auditory acuity to permit adequate perception of instructions or task stimuli, and/or who are of an age falling outside of 18-85.

ATTENTION: Any components removed from a Kinarm Lab will not be supported for any other use.

2.3 Conventions

The user information that accompanies the product uses typographical conventions to assist you in finding and understanding information.

- All procedures are numbered and all sub-procedures are lettered. You must complete the steps in the sequence in which they are presented to ensure success.
- Bulleted lists indicate general information related to a function or procedure. They do not imply a sequential procedure.
- When a task has several approaches or option combinations these are indicated as choices. If applicable, the recommended option is listed first. Each choice completes the associated step.
- Control names and menu items or titles are spelled as they are on the system and they appear in highlighted text.
- Click or select means to move the pointer to an object or menu item and press the left mouse button.
- Right-click means to point at an item and then press the right mouse button.

- When an instruction involves menu-based actions, each menu item appears as **Menu Name**, child menus and options appear in order after the primary menu name. The series of menus and options appear as **File > Subject > Export Record**.

NOTE: Notes bring your attention to information that will help you better understand the product (i.e. for information purposes only).

ATTENTION: Attention notifications highlight information that will help you avoid use errors that are not related to safety.

CAUTION: Cautions highlight hazardous situations that, if not avoided, could result in minor or moderate injury.

WARNING: Warnings highlight hazardous situations that, if not avoided, could result in serious injury or death.

2.4 Audience

This information identifies the audience for the guide. These audience definitions are general in nature.

Operator

This individual runs the Kinarm Standard Tests and the Custom Tasks. They are responsible for cleaning the Kinarm Lab after use.

Servicer

This individual is responsible for routine maintenance and, under the direction of Kinarm, repair/rework the Kinarm Lab as required.

2.5 Related Documentation

The different versions of Kinarm Labs that have been released over the years are identified with different part numbers (PN). Each version has its own unique documentation associated with it, so to ensure that you are using the correct documentation, please refer to [3.1 Identifying Kinarm Lab Versions](#).

Kinarm End-Point Lab (PN 14332) Operator's Guide

This guide describes the steps necessary to operate the Kinarm End-Point Lab™ (PN 14332) made by BKIN Technologies Ltd. (dba Kinarm). It covers the procedures starting with before a subject arrives, to setting up a subject, and then moving a subject through a task from start to

finish lab using Dexterit-E. This guide includes information on creating a Custom Task protocol. It also includes information on maintaining the lab.

Dexterit-E User's Guide

This guide describes the capabilities of the Dexterit-E™ software. The Dexterit-E software is used in conjunction with a Kinarm lab. Some basic information is shared in the lab's operator's guide to simplify operation. It is important to be familiar with the contents of this guide to understand the full range of features and functionality available in Dexterit-E.

This guide includes the user interface, upgrading the software, selecting and backing up databases, managing subject data, and running tasks. This guide describes the Kinarm Standard Tests and discusses creating Custom Tasks, task protocols, and analysis.

The guide includes information on calibrating and verifying the associated lab.

Other Available Guides

- **Adjustable Height Configuration Addendum for Hardware and Operator's Guides** - Provides a description of the hardware for the adjustable height configuration of the Kinarm End-Point Lab™.
- **Creating Task Programs** - Introduces the concept of Task Programs, the basic code structure of Task Programs, and demonstrates these with numerous examples.
- **Dexterit-E Explorer Users' Guide** - Provides instructions on using the Kinarm Dexterit-E Explorer™ program for viewing data collected in Dexterit-E.
- **Eyelink 1000+ User Manual** - Provides more detailed instructions on using the Eyelink 1000+ software, which is a core component of the Kinarm Gaze-Tracker.
- **Kinarm Analysis Scripts** - Provides a brief description of the various functions that can be used to load, manage, and analyze Kinarm data in MATLAB™.
- **Kinarm Standard Tests Summary** - Describes the Kinarm Standard Tests™ (KST) available as part of the Dexterit-E™ software usable on the Kinarm Lab.
- **Task Development Computer Setup Guide** - An excerpt from the Dexterit-E Addendum - Creating Task Programs that provides all necessary information for setting up the development environment to create Custom Task Programs for Dexterit-E.

2.6 Warnings and Cautions

Custom Task Creation

CAUTION: Ensure the emergency stop button is readily accessible and/or keep a firm grip on the robot(s) when testing newly edited Custom Task Programs. Applying new loads can result in unexpected behaviour such as excessive forces and/or unstable loads. Care must

be taken to reduce the risk of injury to the subject or operator, and/or damage to the Kinarm Lab.

Electrical and Heat Safety

WARNING: To avoid the risk of electric shock, the Kinarm Lab must be connected to a supply mains with protective earth.

WARNING: To reduce the risk of electric shock, do not connect anything that is not part of the Kinarm Lab to the Power Console.

WARNING: To avoid the risk of electric shock, do not remove enclosure panels.

WARNING: To avoid the risk of electric shock, keep water and other liquids away from the Kinarm.

Hygiene

CAUTION: Clean all components of the system that come into contact with subjects between subjects to reduce the risk of pathogen transfer.

General Lab Setup and Use

CAUTION: Do not allow bystanders within the robot workspace. Ensuring that bystanders remain outside of the robot workspace at all times reduces the risk of injury.

CAUTION: Do not alter or remove any hardware from the lab. Hardware that requires the use of a tool to remove or adjust should not be detached or repositioned unless under the direction of Kinarm Support. Leaving the hardware as originally installed reduces the risk of subject injury. This includes ensuring the safety stops are not removed.

CAUTION: Do not install unsupported software/OS on the Kinarm Lab computers. Avoid running any program simultaneously with Dexter-E (such as virus scans, web browsers, copying files, email, watching movies, etc.) to reduce the risk of degrading system performance.

CAUTION: Do not overload the Power Rack drawer. Keeping the total mass of items in the drawer to a maximum of 5 kg reduces the risk of the Power Rack tipping.

2.7 Updates and Upgrades

Kinarm is committed to innovation and continued improvement. When upgrades that consist of hardware or software improvements are released, updated user documentation will accompany those upgrades.

Software downloads for your Kinarm Lab can be found at:

- <https://kinarm.com/learning-support/software-downloads/>

Documentation for your Kinarm Lab can be found at:

- <https://kinarm.com/learning-support/user-guides-documentation/>

2.8 Customer Service

If you encounter problems with your Kinarm Lab or have questions about the user documentation, contact Kinarm Support at:

- support@kinarm.com

If applicable, generate and include a log file from Dexterit-E (**Help > Generate Log**; see the Dexterit-E User Guide for more details).

3 In This Guide

Welcome to the Kinarm End-Point Lab (PN 14332) Hardware Guide. This guide describes the various hardware components of the Kinarm End-Point Lab™ with part number 14332. Various sections of this guide include: installing and setting up the Kinarm End-Point Lab and an overview of all hardware components of the Kinarm End-Point Lab. For instructions on operating the Kinarm End-Point Lab, refer to the Kinarm Lab Operator Guide. For more details about using the Kinarm End-Point Lab with Dexterit-E™, the software designed to run the Kinarm End-Point Lab, refer to the Dexterit-E User Guide. For more details about using Kinarm Standard Tests™, refer to the Dexterit-E User Guide or the Kinarm Standard Tests Summary.

ATTENTION: This version of the hardware guide is applicable to Kinarm End-Point Lab 14332 revisions that contain Power Console PN 15050 and an Audio Visual Controller (AV Controller) first available in 2025.

3.1 Identifying Kinarm Lab Versions

The different versions of the Kinarm Exoskeleton Lab™ and Kinarm End-Point Lab™ that have been released over the years are identified with different part numbers (PN). Although overall functionality is the same for different versions, changes in hardware design have required some changes in appearance and operation. This section provides information that can be used to identify which revision of a Kinarm Lab you have.

Newer Kinarm Labs have labels indicating both a part number (PN) and a serial number (SN). If present, the PN label can be used to identify your Kinarm version. If not, then a SN Label may be of assistance - the first two digits of a SN Label indicate the year that the Kinarm was manufactured.

PN 10288. The first version of the Kinarm End-point Lab was released in 2009, and was in production until 2020. This version did not have a PN label for the Kinarm Lab, but each Kinarm Robot had its own PN label: PN 10212 for the right robot and PN 10213 for the left robot. If your Kinarm End-Point robots have individual Kinarm robot PN labels 10212 and 10213 and have a SN label starting with 20 or less, then the PN for your Kinarm Lab is PN 10288.

PN 14332. The second version of the Kinarm End-point Lab was released in 2021. It has a PN label on the back of the right Kinarm Robot indicating PN 14332.

ATTENTION: In late 2025, a new version of the Kinarm End-Point Lab PN 14332 was released, containing Power Console PN 15050 and an AV Controller. Rev 2 of the hardware guide applies to this new revision. Older Kinarm End-Point Labs PN 14332 should continue to use Rev 1 of the hardware guide as the main reference, but can reference Rev 2 for information on features not covered by Rev 1.

4 The Kinarm Lab

The Kinarm Lab is a complete research lab for studying brain function that is built around the high performance Kinarm Robot.

4.1 How it Works

The Kinarm End-Point robot is a stiff, graspable robot that can monitor and manipulate arm motion in the horizontal plane. The Kinarm End-Point lab is a modular system that includes either one or two robots for either unilateral or simultaneous left and right-handed arm manipulation. The Kinarm End-Point Lab integrates a 2D virtual target presentation system and comes complete with Dexterit-E™ software for controlling behavioural paradigms and recording data. A Kinarm Gaze-Tracker can be added to the Lab to provide gaze-tracking capability to the Kinarm End-Point Lab.

4.2 Benefits

- Provides a general platform to perform a broad range of sensorimotor and cognitive tasks.
- Can create highly complex mechanical environments by applying loads to the hand.
- Generates a broad range of quantitative information on motor performance.
- The use of two Kinarm End-Point robots permits comparison of inter-arm performance as well as the study bi-manual coordination.
- Precisely calibrated for repeatable measurements.
- Kinarm End-Point Labs are modular in design, providing a cost-effective way to build a lab over time. Options can be selected that will fit each researcher's needs. Typical solutions range from a single, standalone robot to a complete lab comprising two robots with high resolution encoders and built-in force sensors, integrated virtual reality and data acquisition systems, and gaze-tracking.
- Includes Dexterit-E, Kinarm's behavioural control and data acquisition software, for a complete, ready to use system:
 - Dexterit-E has easy-to-use Windows™ based interface.
 - Dexterit-E task control is through a computer running a precise real-time operating system.
 - Dexterit-E provides Standard Task Programs which can be used immediately for data collection, analysis and report generation.
 - Dexterit-E permits researchers to create unique, customized behavioural Task Programs.
 - Dexterit-E comes with a Task Development Kit and samples of pre-developed tasks to assist with rapid Custom Task Program development.

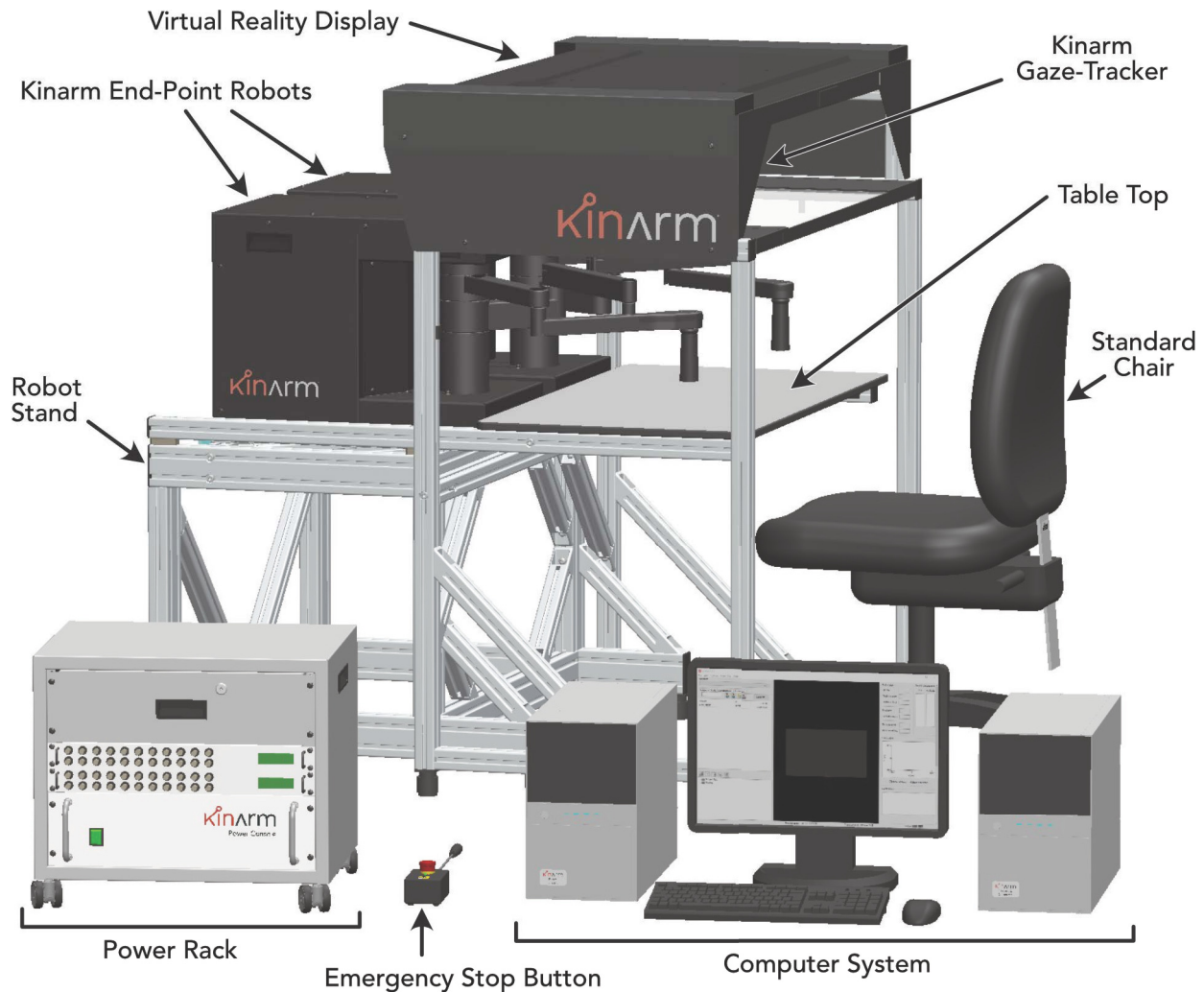
5 Components of the Kinarm Lab

Kinarm End-Point Labs are modular in design and therefore can come with a number of optional components. This modular nature allows basic Kinarm Labs to be upgraded in the field, by adding the necessary modules. This chapter of the Kinarm End-Point Lab Hardware Guide provides an overview of the standard and optional components that can be a part of a Kinarm End-Point Lab. Please refer to the Dexterit-E User Guide and the Kinarm Lab Operator Guide for instructions on use of the various lab components.

5.1 Main Components

The main components and options for the Kinarm End-Point Lab are described below and illustrated in [Figure: 5.1](#), with further details on subcomponents and modules in the following sections.

Figure 5.1: Main components of the Kinarm End-Point Lab



- **Control System** - The control system to run Dexterit-E is based on two computers and a controller. The Robot Computer runs a real-time OS for controlling the Kinarm End-Point Lab, the AV Controller (not shown) drives the subject display, while the Dexterit-E Computer provides a Windows-based user interface. The Robot Computer may contain various PCI-based cards for interfacing with hardware, such as a DAQ card for collecting analog inputs.
- **Deluxe Chair (Optional, Not Shown)** - A battery-operated chair with adjustable height is provided for the subject. The Deluxe Chair does not spin or roll when the brakes are applied so that the subject remains in the same position while using the robot. The Deluxe Chair comes with adjustable armrests, footrest, backrest, and can include a subject support harness and hip belt.
- **Emergency Stop Button** - The emergency stop button allows instant disabling of all motors in the Kinarm End-Point Lab by depressing the button.
- **Kinarm End-Point Robot** - The Kinarm End-Point Robot is a motorized mechanism that allows manipulation of the arm in the horizontal plane. One or two Kinarm robots may be

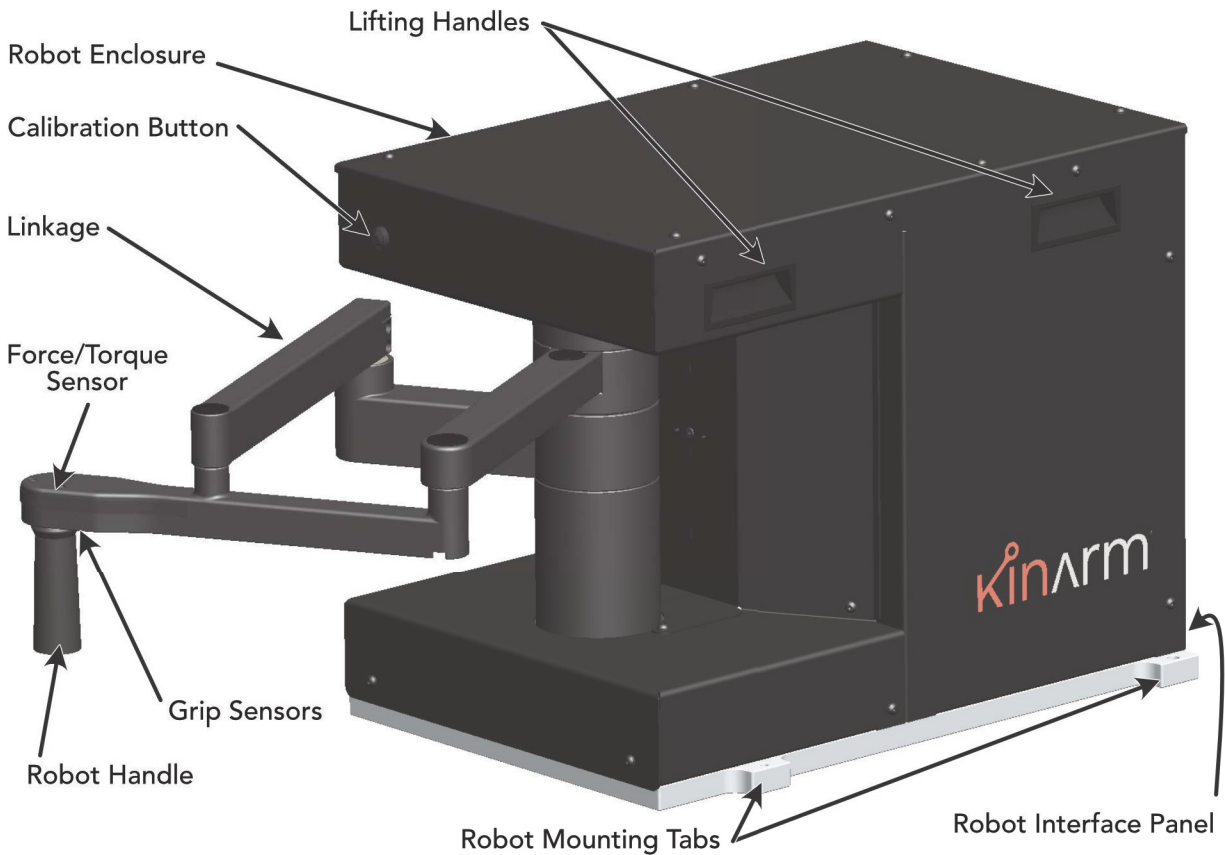
used for unilateral or bilateral operation respectively. Absolute secondary encoders provide high-resolution position information and optional 6-DOF force/torque sensors can be integrated into the handle of each robot.

- **Kinarm Gaze-Tracker (Optional)** - The optional Kinarm Gaze-Tracker provides integrated gaze-tracking capability within the Virtual Reality Display.
- **Power Rack** - This 19" rack contains several rack-mounted components. The Power Console is the interface between the robots and the Robot Computer, providing power to the robots and a connection for the emergency stop button. The rack also houses the BNC breakout panels for the optional DAQ system. Additionally, a built-in drawer is provided for storage of accessories.
- **Robot Stand** - The robot stand is a stiff support structure for the Kinarm End-Point robots. The stand locates both robots relative to each other and raises the handles of the robots to a comfortable desktop height.
- **Standard Chair** - A chair with adjustable height is provided for the subject. The chair does not spin or roll when the brakes are applied so that the subject remains in the same position while using the robot
- **Tabletop** - The tabletop provides a surface for the Kinarm End-Point Lab to ensures that the subject's body is kept outside the robot's workspace
- **Vertical Display (Optional, Not Shown)** - The vertical display option replaces the Virtual Reality Display and provides an upright subject display mounted on top of the Kinarm robots.
- **Virtual Reality Display** - The virtual reality display provides a 2-dimensional virtual reality setup for the subject to interact with. It includes a display device and semi-transparent mirror to create perception of the image in the plane of the robot. A vision blocker to optionally prevents the subject from viewing the robot or their own arms.

5.2 Kinarm Robot

The Kinarm End-Point Robot is the central component of each Kinarm Lab. [Figure: 5.2](#) shows the components of the robot.

Figure 5.2: Kinarm End-Point Robot Components

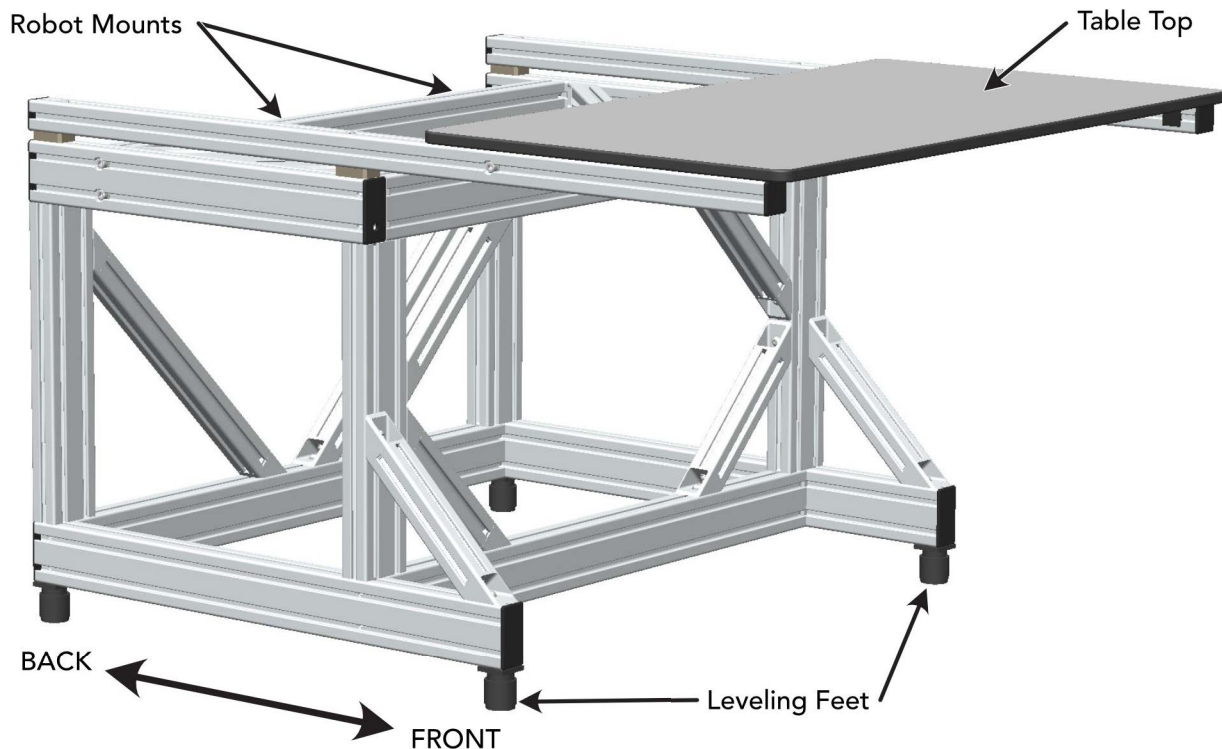


- **Absolute Encoders (Not Visible)** - The absolute high-resolution encoders provide accurate position information and remove the need for daily calibration. They are housed within the robot enclosure.
- **Calibration Button** - The calibration button is used to indicate to Dexter-E when the robot is in position during factory calibration. The button is located on the top left corner of the front panel of the robot enclosure for both right and left handed robots.
- **Force/Torque Sensor (Optional)** - A 6-DOF force/torque sensor may be integrated into the handle of the Kinarm End-Point robot. The wire is routed along the underside of the robot links.
- **Grip Sensors** - Proximity sensors are embedded above the handle to indicate to the robot whether the handle is being grasped. When the robot is enabled, it integrates an active braking feature which prevents sudden movement of the linkage when the handle is released. Active braking is disabled when the handle is grasped, and enabled when the subject releases the handle.
 - **NOTE:** The grips sensors do not measure the grip force.
- **Joint Limits (Not Visible)** - The robot has several internal end stops which limit the movement of the robot to a safe range of motion. The range of motion of a unilateral robot is slightly different than a bilateral robot to accommodate the available workspace.

- **Lifting Handles** - The robot enclosure integrates four lifting handles, two on each side of the robot to facilitate lifting and positioning of the robot.
- **Linkage** - The Kinarm End-Point Robot is a four-bar linkage mechanism. The four links are designated as Link L1, Link L2, Link L3, and Link L4 as described in [9.3 Robot Link Definitions and Dimensions](#).
- **Robot Enclosure** - The robot enclosure covers the internal mechanical and electrical components.
- **Robot Handle** - The robot handle freely rotates about the end of link L2. The handle is mounted to an optional force/torque sensor, and consists of a comfortable tapered handle grip to prevent slipping. A version of the handle with wrist support is optionally available. Users can also create a custom handle (see [10.3 Custom Handles](#))
- **Robot Interface Panel** - The robot interface panel is located on the back of the robot enclosure, and it is where connections for power and communication are made. Additionally, the interface panel includes status LEDs and digital inputs and outputs.
- **Robot Mounting Tabs** - The robot has four tabs on its base which are used for mounting to the robot stand.

5.3 Robot Stand

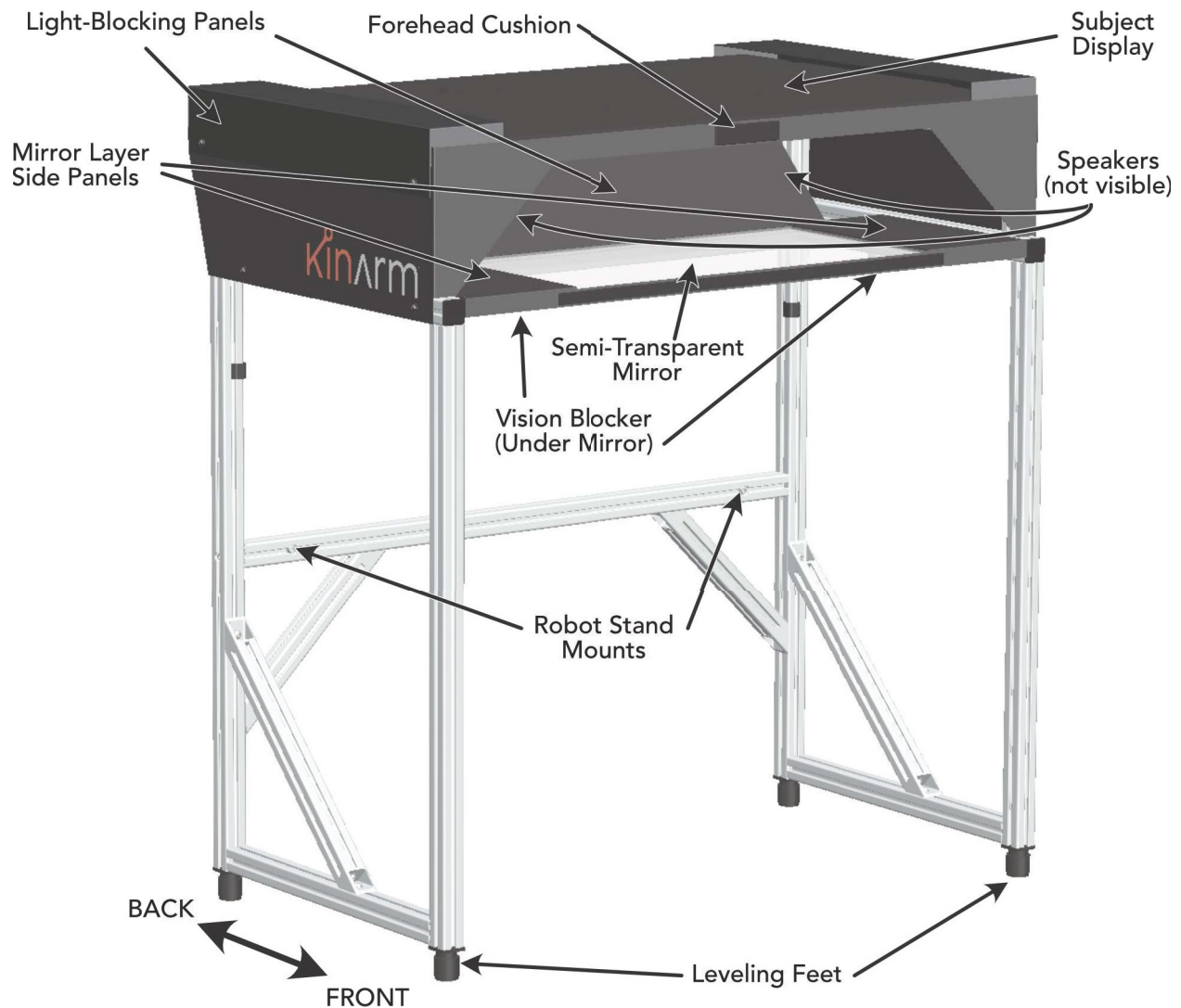
The robot stand is a stiff support structure on which one or two Kinarm End-Point robots can be mounted. [Figure: 5.3](#) shows main components of the stand.

Figure 5.3: Robot Stand Components

- **Leveling Feet** - The four feet of the robot stand are adjustable in height for leveling purposes.
- **Robot Mounts** - The robot stand can accept either one or two Kinarm End-Point robots. Each robot is attached to the robot stand using four screws.
- **Table Top** - The table top is a desktop surface mounted underneath the robot handles. It is required to safely maintain subject position, preventing the subject's body from moving into the path of the robots.

5.4 Virtual Reality Display

The virtual reality display is a 2-dimensional virtual reality display system attached to a support structure. [Figure: 5.4](#) shows the main components of the virtual reality display.

Figure 5.4: Virtual Reality Display Components

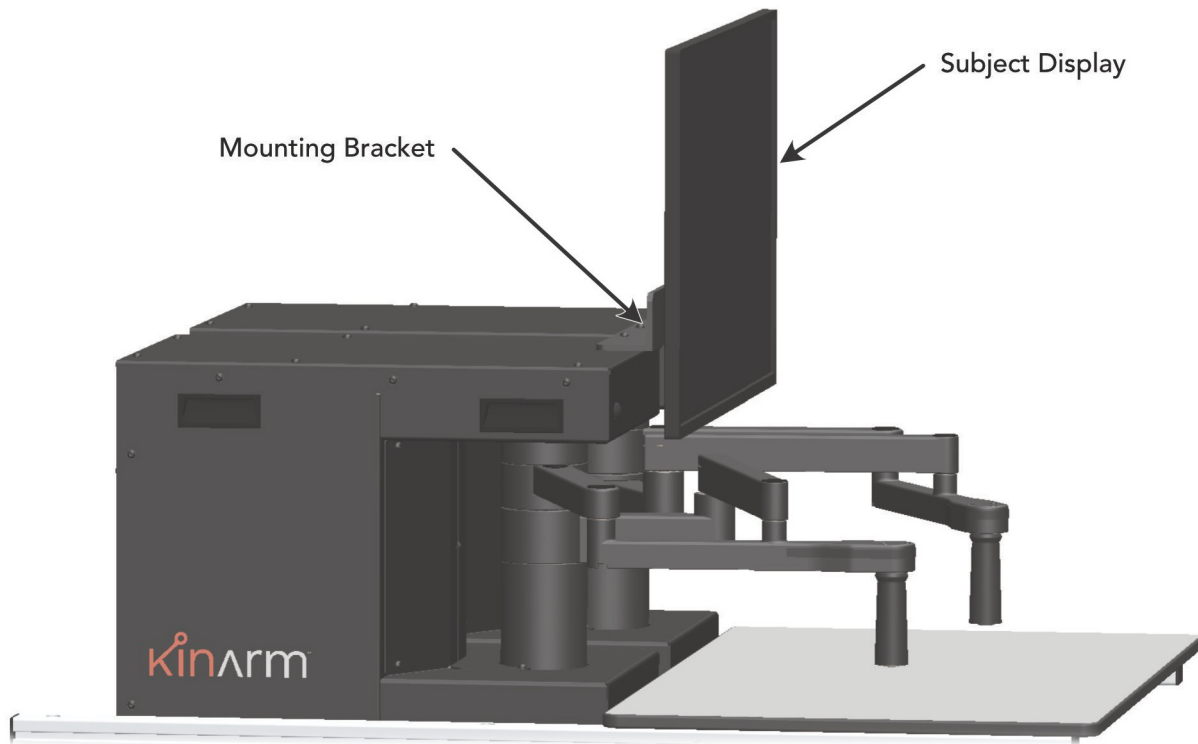
- **Forehead Cushion** - A pad on which a subject can rest their forehead during a session. The forehead cushion is also used aid with subject positioning.
- **Leveling Feet** - The four feet of the virtual reality display are adjustable in height for leveling purposes.
- **Light-Blocking Panels** - Panels mounted around the sides and back of the display prevent room lighting from interfering with the displayed images.
- **Mirror Layer Side Panels** - These side panels on the mirror layer block vision through the sides of the mirror layer and hold the mirror in place.
- **Robot Stand Mounts** - The virtual reality display is fixed to the robot stand using two screws.
- **Semi-Transparent Mirror** - The semi-transparent mirror is located in a horizontal layer underneath the video display. Because it is semi-transparent, the subject can see their

arms while simultaneously viewing a reflection of the subject display images, such that they appear in the same plane as the arm.

- **Speakers (Not Visible)** - The speakers provide crisp, clear stereo audio to the subject, for use with providing task instructions and/or audio elements in custom tasks.
- **Subject Display** - This flat-screen monitor provides the video display for the virtual reality system. Visual stimuli that the subject will view, via a reflection in the semi-transparent mirror, are displayed by this screen. The display can be upgraded from the standard 60 Hz model to a 120 Hz model for low latency visual presentation.
- **Vision-Blocking Bib (Not Shown)** - The vision blocking bib fits around the subject's neck and to the front of the semi-transparent mirror layer to block the view of the subject's arms and hands.
- **Vision Blocker** - The vision blocker can slide inwards from both sides of the semi-transparent mirror layer to block the view of the subject's arms and hands, or slide outwards to expose the subject's arms and hands. The vision blocker does not block any images displayed by the display.

5.5 Vertical Display

An alternative to the virtual reality display is the vertical display, which is an upright subject display located above the robot linkage(s). [Figure: 5.5](#) shows the components of the display.

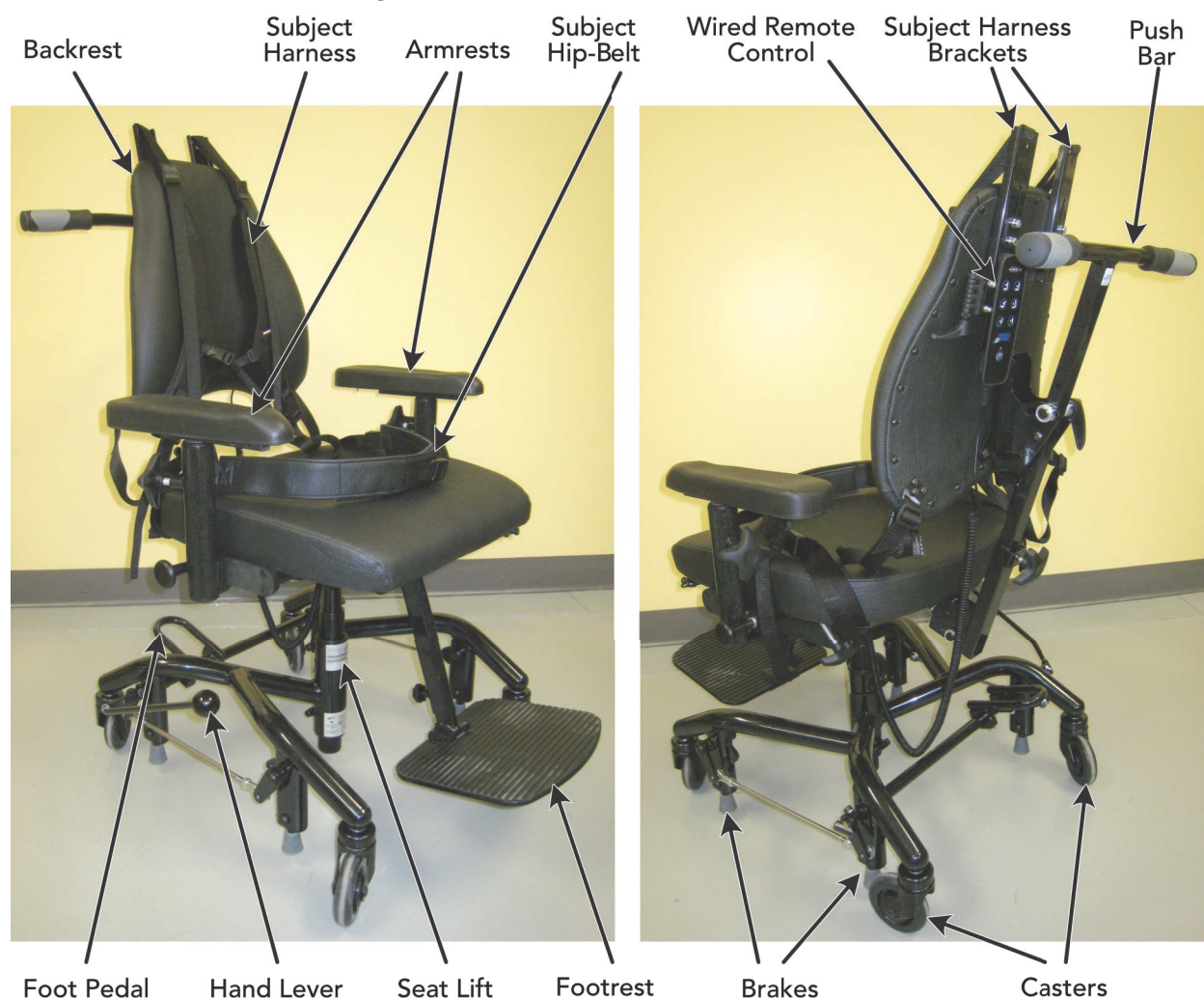
Figure 5.5: Vertical Display Components

- **Mounting Bracket** - The vertical display is mounted directly to the top of the robot enclosure(s) using custom robot mount brackets and the monitor's VESA display mount. The display can be centred on either one robot or a pair of robots.
- **Subject Display** - The video display is a ~32" flat screen monitor.

5.6 Deluxe Chair

The optional Deluxe Chair is a configuration of the REAL® 9000 Plus EL manufactured by Mercado Medic (Lidingö, Stockholm, Sweden) and comes with a more clinical feature set as compared with the Standard Chair. The Deluxe Chair is designed to accommodate a range of subject sizes, up to a maximum of 330 lbs (150 kg). [Figure: 5.6](#) shows some overall images of the chair with all of the main components labeled.

Figure 5.6: Deluxe Chair Components



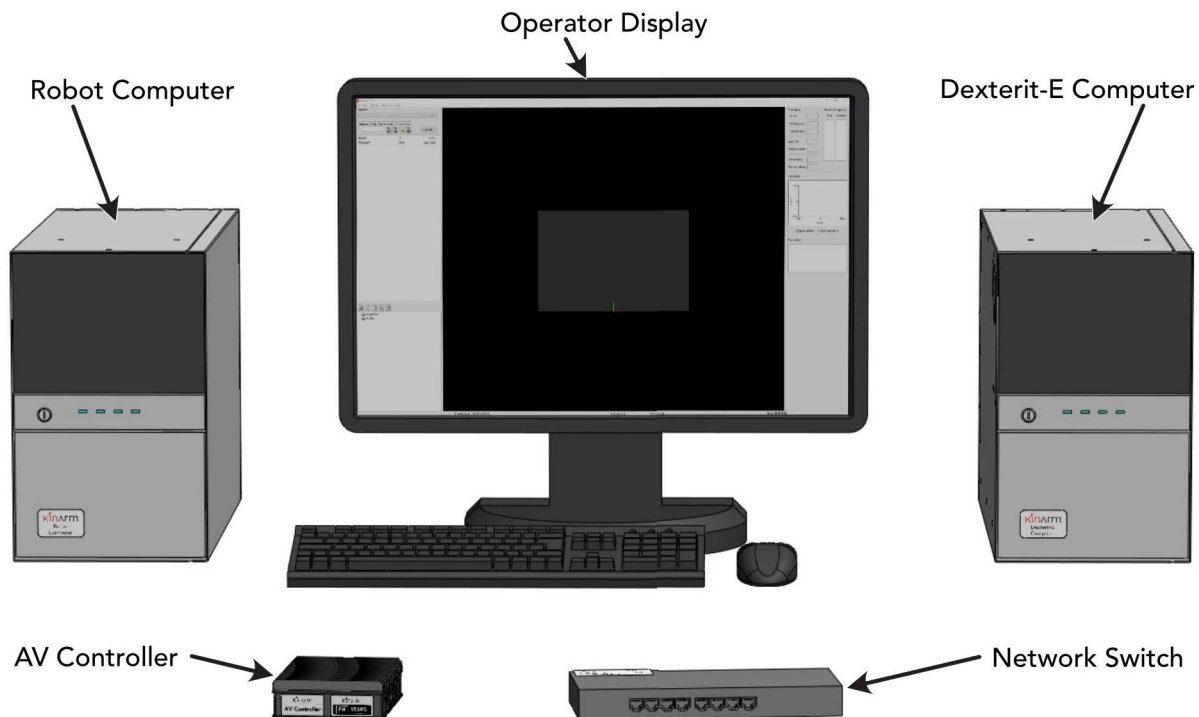
- **Armrests** - The armrests are used to assist with subject transfer. They can be manually adjusted in height as well as in the medial-lateral direction if needed. The armrests fold backwards to allow side-entry to the chair, and to prevent interference with the subject's arms during exams.
- **Backrest** - The backrest can be manually adjusted in three ways. It can be adjusted in height and angle, and the backrest pad itself can be pivoted. The backrest pad is shaped to avoid interference with the subject's arms during an exam.
- **Battery/Controller Box (Not Visible)** - The chair's controller and battery are enclosed in a box mounted underneath the chair.
- **Brakes** - The chair comes with a 4-point brake system that can be engaged and disengaged by the subject and/or the operator. There are two ways to use the brakes:
 - **Hand Lever** - The chair brakes can be engaged and disengaged by the subject or operator using the hand lever. The hand lever can be installed on the right (default) or left side of the chair.

- **Foot Pedal** - The foot pedal can be accessed by the operator from the back of the chair.
- **Casters** - The chair has four swivel casters. As a result, the chair can be moved in any direction.
- **Charger (Not Shown)** - The chair is battery operated, and therefore must be charged between uses in order to have sufficient power to operate the height adjustment and to maintain battery health. The chair specifically requires the included charger to properly charge the battery.
- **Charging Port (Not Visible)** - The charger can be plugged into the chair at the charging port, located under the seat at the front right corner, or on the left armrest.
- **ATTENTION:** The height of the seat cannot be adjusted while the battery is charging.
- **Footrest** - The footrest is a single platform centrally mounted on the chair that can be used to support the feet of the subject. The footrest height can be adjusted to accommodate subjects of different height. The footrest can be folded away during transfer and when not in use.
- **Push Bar** - On the back of the chair is a push bar that can be used by the operator to move the chair into the desired position.
- **Seat Lift** - The height of the seat is adjusted with an electronic actuator, located in the base of the chair.
- **Subject Harness** - An H-strap harness is attached to the backrest to assist with maintaining subject torso position during exams. It can also be used in combination with the belt to provide some postural support.
- **Subject Harness Brackets** - The harness brackets located at the top of the backrest can be adjusted in height (5 mm hex key is provided with the chair) to accommodate subjects of different height. If necessary, the brackets can also be pivoted outwards to provide a wider support.
- **Subject Hip Belt** - A hip-belt is attached to the chair to assist with subject positioning. It can also be used in combination with the subject harness to provide some postural support.
- **Wired Remote Control** - The seat lift is controlled with a hand-held device. The wired remote control is connected to the chair by a coiled cable, so it can be moved around with the operator, but cannot be removed from the chair. Only up and down seat height adjustments are possible.

5.7 Control System

The control system contains all components necessary to run the Kinarm Lab.

Figure 5.7: Control System Components



- **Audio Visual (AV) Controller** - This device controls the subject display and Dexterit-E audio output. The AV Controller is mounted out of sight in the back of the Power Rack.
- **DAQ PCI Card (Optional, Not Visible)** - The DAQ option comes with a National Instruments PCI-6229 data acquisition card which includes up to 32 analog inputs for recording analog data, 4 analog outputs and 46 general purpose digital I/O. This card is installed in the Robot Computer, and is controlled directly through Dexterit-E.
- **Dexterit-E (Not Visible)** - The software used to control the Kinarm End-Point Lab and to collect and organize the data. It runs on the Dexterit-E Computer.
- **Dexterit-E Computer** - This computer runs the Windows operating system, and it controls the Robot Computer and AV Controller, and stores data.
- **Dexterit-E LAN (Not Shown)** - The Robot Computer, Dexterit-E Computer, AV Controller and various optional components communicate via a dedicated local area network (LAN) named the Dexterit-E LAN. The Dexterit-E LAN includes a network switch.
- **EtherCAT Network (Not Shown)** - Real-time control of the robot is achieved through a dedicated EtherCAT based network. This network is separate from the Dexterit-E LAN and comprises the Robot Computer, Power Console and the servoamplifiers of the Kinarm robots.
- **Kinarm Standard Tests (Optional, Not Visible)** - This optional component of Dexterit-E is a suite of standardized tests, including pre-defined tasks for data collection and associated analysis and reporting capabilities against normative data. Please refer to the Dexterit-E User Guide for more details.
- **Network Switch** - This switch is required to connect the Dexterit-E LAN for all configurations, including when either a third computer (i.e. gaze-tracking) or force/torque

sensors are used. The network switch provides power over Ethernet (PoE) to the force/torque sensors. The network switch is mounted out of sight in the back of the power rack.

- **Operator Display** - The display through which the operator sees and interacts with Dexterit-E.
- **Robot Computer** - This computer runs the Simulink Real-Time operating system, and it controls the Kinarm robots through a dedicated EtherCAT network (separate from the Dexterit-E LAN). The Robot Computer contains data acquisition cards, if applicable.
- **Robot Computer Video Output (Not Visible)** - The video from the Robot Computer can be viewed in different ways, depending on the configuration of the Kinarm Lab. The typical configuration is a screen capture device that allows the video to be streamed to the Dexterit-E computer. The software for this setup makes it easy to take screen shots or video recordings of the display. Alternatively, a user-supplied display may be connected directly to the Robot Computer.
- **Subject Display (Not Shown)** - The display that shows the subject visual targets and cues. The subject display is part of the Virtual Reality Display ([Figure: 5.4](#)) or the Vertical Display ([Figure: 5.5](#)).

5.8 Power Rack

The Power Rack is a rollable rack that contains several 19" rack-mount components. [Figure: 5.8](#) shows the main components of the Power Rack.

Figure 5.8: Power Rack Components



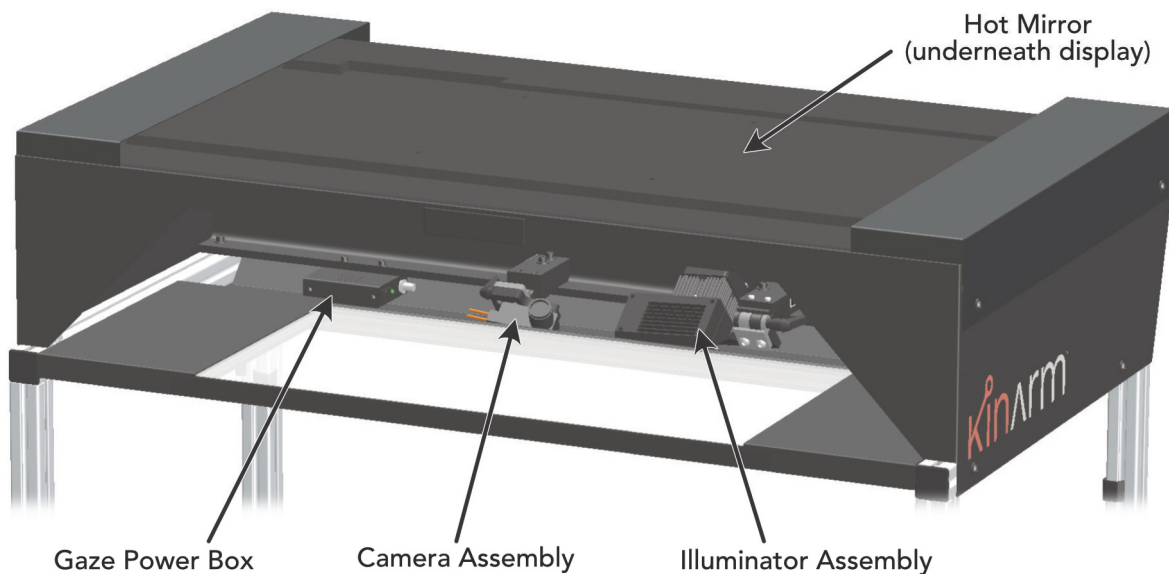
- **AV Controller (Not Shown)** - Functionally, the AV Controller is part of the control system (see [5.7 Control System](#)), but it is mounted in the back of the Power Rack.

- **BNC Breakout Panels (Optional)** - The inputs and outputs of the optional DAQ system are accessed from one or two National Instruments BNC-2090A breakout panels. Each panel provides access to half of the channels.
- **Network Switch (Not Shown)** - Functionally, the network switch is part of the control system (see [5.7 Control System](#)), but it is mounted in the back of the Power Rack.
- **Power Console** - This console is the power and communication centre for the Kinarm Lab. The Power Console provides power to all components of the Kinarm Lab, and is the interface for the emergency stop button as well as the communication link between the robots and the Robot Computer.
- **Rack** - The rack provides a frame to support the various 19" rack-mount components of the Power Rack. There is also a panel on the back of the rack (not shown) on which the network switch and AV Controller are mounted (see [Section 5.7 Control System](#)). If desired, the computers can be set up on top of the rack. The rack is mounted on swivel casters with brakes that facilitate positioning and access to the components.
- **Storage Drawer** - The drawer provides storage space for the various accessories belonging to the Kinarm End-Point Lab. The drawer typically is used to store the calibration bar and a selection of tools necessary for Lab installation. The drawer may also be used to store other accessories including the microfibre cloth, vision-blocking bib and the optional handles and hand straps.

5.9 Kinarm Gaze-Tracker

The Kinarm Lab optionally comes with the Kinarm Gaze-Tracker ([Figure: 5.9](#)), which requires the Virtual Reality Display. The components of the Kinarm Gaze-Tracker are detailed below.

Figure 5.9: Kinarm Gaze-Tracker Components



- **Camera Assembly** - This assembly mounts the camera head in the correct position to obtain a clear view of the eye across the majority of the robot workspace.
- **Gaze-Tracker Computer System (Not Shown)** - This computer runs the software that collects the gaze data. It comprises a computer, with associated cards and camera base box, as well as a monitor, keyboard, and mouse.
- **Gaze Power Box** - This box consolidates all the power connections for the components mounted within the Virtual Reality Display.
- **Hot Mirror (Not Visible)** - This mirror is located under the subject display screen and is required for the Kinarm Gaze-Tracker to properly track the eye. The hot mirror reflects infrared light while simultaneously transmitting visible light.
- **Illuminator Assembly** - This assembly contains an array of infrared LEDs, and an associated mounting system, which provide illumination of the eye and allows the camera to track the gaze of the eye.
- **Target Stickers (Not Shown)** - Target stickers are placed on the subject's head to assist with the head-free operation of the Kinarm Gaze-Tracker.

6 Site Readiness

The information provided in this chapter should be used as reference to plan the installation or move of a Kinarm Lab. It includes an overview of the minimum requirements for the room or space in which the Kinarm Lab is to be installed. Guidance provided in this chapter will help to ensure that the space is prepared for a successful installation. Our sales and support team is happy to consult with you prior to any installation or move.

Additional documentation is available at:

- <https://kinarm.com/learning-support/installing-maintaining-your-kinarm-lab/>

6.1 Shipping and Receiving

Kinarm End-Point Labs typically ship in 2 large crates. Each crate has dimensions 1.3 m x 1.1 m x 1.55 m, with a combined total mass of approximately 500-600 kg depending on the configuration. Kinarm Sales will work with the customer to prepare a receiving plan.

6.2 Environmental

The Kinarm Lab must be installed indoors, and is intended to be used in the following environmental conditions:

- Operating temperature range:
 - 10 to 40 °C (excluding Kinarm Gaze-Tracker)
 - 10 to 30 °C (Kinarm Gaze-Tracker);
- Storage temperature range: -20 to 60 °C;
- Operating humidity range: 20-80%;
- Storage humidity range: 10-85%
- Operating altitude: up to 2000 m.

6.3 Electrical

The Kinarm Lab requires a single outlet with the following characteristics:

- dedicated circuit (i.e. circuit not shared with other outlets/equipment);
- supply providing either:
 - 20 A @ 100-120 VAC, 50/60 Hz, or
 - 10 A @ 200-240 VAC, 50/60 Hz;
- located within 2 m of the Kinarm Lab's Power Rack;

- proper grounding (i.e. protective earth).

NOTE: Surge protection is not needed because it is built into the Kinarm's Power Console.

6.4 Internet

We recommend an Ethernet port be available for communications with the external internet to assist customer support and upgrades to Dexterit-E software. The Kinarm Lab does not require an internet connection for its function.

ATTENTION: As noted in the Administration Chapter of the Dexterit-E User Guide, installation of most software and many configuration changes to Windows on the Dexterit-E computer are not supported for Kinarm Labs. This restriction may contravene the requirements of your local IT department, and therefore may prevent you from being able to connect the Dexterit-E computer directly to the internet. Prior to the arrival of your Kinarm, please consult with your local IT department to determine if you will be allowed to connect the Dexterit-E computer to the internet.

6.5 Lighting

For rooms with bright overhead lights and/or with windows, we recommend modifying the room to allow control of the ambient light. This might include installing new switches, dimmers or baffles for overhead lights, and/or installing blinds or curtains on windows.

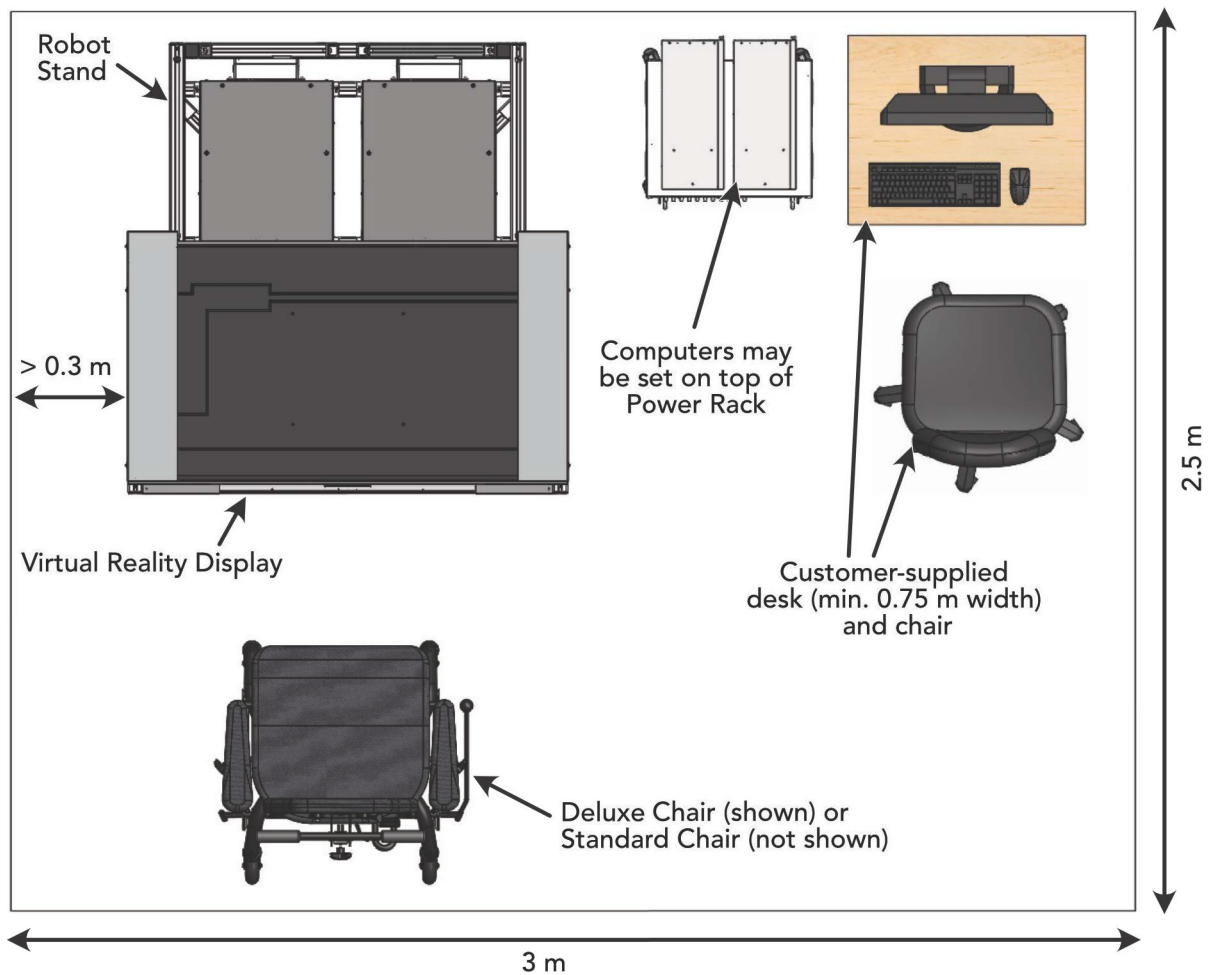
6.6 Flooring

The End-Point chair moves well on either hard carpet (i.e. with minimal underlay padding) or laminate/rolled/solid-surface flooring. However, in low-humidity regions, carpet is not recommended due to static build-up associated with carpet.

6.7 Kinarm Lab Placement

The minimum required space for the Kinarm End-Point Lab is 2.5 m x 3 m, however at least 3 m x 3.5 m is recommended for clinical researchers working with impaired populations that may require a transfer from a wheelchair. The suggested layout for a typical room is shown in [Figure: 6.1](#).

Figure 6.1: Suggested Kinarm End-Point Lab Layout



6.7.1 Dimensions and Spacing

- The combined Virtual Reality Display and Robot Stand is 1.2 m (width) x 1.2 m (depth).
- The back of the Robot Stand is typically placed against one wall.
- At least 0.3 m space should be available on each side of the Virtual Reality Display.
ATTENTION: This is the minimum recommended space to ensure access to the back of the robots (i.e. status indicators, DIO) and to allow space for subject transfer.
- A desk or worktable for the operator is not provided with the Kinarm Lab.
 - The customer should provide a suitable desk or table to hold the operator display, keyboard and mouse (min. 0.75 m width).
 - If the Kinarm Gaze-Tracker is included, a larger desk is recommended to fit a second display, keyboard and mouse (min 1.2 m width).

- The Dexterit-E and Robot Computers can both be placed on top of the Power Rack (if using nominal desk size recommended) or on top of a much-wider desk.

NOTE: If the computers are placed on top of the Power Rack, the combination may not fit under a typical desk. The combined height is approximately 0.8 m, whereas the height of the rack alone is approximately 0.47 m. The width of the Power Rack is 0.55 m.

6.7.2 Component Arrangement

The various components of the Kinarm Lab are connected via cables and so have maximum spacing restrictions due to cable lengths. The distances listed below are nominally based on centre-to-centre spacing of components, and should be interpreted as approximate guidelines only.

- The control system can be located on either side of the Virtual Reality Display.
- The Power Rack must be within 3 m of the Kinarm Robots and within 2 m of the supply outlet.
- The control system must be within 3 m of the Power Rack.

NOTE: If the optional DAQ with BNC Breakout is included in the Power Rack, the Robot Computer must be within 1.5 m of the Power Rack

- The operator display, keyboard and mouse must be within 3 m of the Dexterit-E Computer.
- The components of the Virtual Reality Display including the subject display, speaker amplifier and Kinarm Gaze-Tracker components must be within 4.5 m of the Power Rack.
- The components of the Virtual Reality Display including the subject display and speaker amplifier must be within 4.5 m of the Dexterit-E Computer.

6.8 Custom Task Program Development

To create custom tasks, the user must set up a custom task programming environment. If on-site training is purchased, this environment must be set up prior to the arrival of the Kinarm Trainer. A license to specific tools from MathWorks is required. Often, the toolboxes may be incremental to the standard MATLAB license your institution holds. See the Task Development Computer Setup Guide for details on which toolboxes are required.

NOTE: A MATLAB license is not required by the Kinarm Lab to operate the Kinarm. It is only required for Custom Task development.

6.9 Wearing Eyeglasses with the Kinarm Gaze-Tracker

For eyeglass-wearing subjects, it is recommended that the customer purchase a pair of glasses with interchangeable lenses that are suited to gaze tracking. For example, contact Solo Bambini (<https://solobambini.com/>) for a custom solution.

Additionally, the customer may wish to purchase an instrument for measuring the strength of the subject's lenses as not everyone will know what their lens prescription is.

7 Unpack and Assemble

This chapter of the guide describes how to unpack and install the Kinarm Lab. The order of installation is important, so installation should proceed as outlined in the guide. Some of the installation steps may involve optional components; skip any steps that do not apply to your configuration. Your shipment will include a packing list to facilitate tracking crate contents once you have removed the items from the packaging.

Before assembling the Kinarm Lab it is a good idea to decide where the final desired location of the lab is to be. Considerations for lab placement are reviewed in [6 Site Readiness](#).

7.1 Inspect and Unpack the Crates

A Kinarm Lab is shipped in one or more crates. To ensure that any potential problems are properly covered by warranty, before opening the crates they should be inspected for any damage that might have occurred during shipping.

1. Inspect the crates for damage.
 - If there is visible damage, take photos of it and contact Kinarm for directions on how to proceed. Do not continue further until instructed by Kinarm.
 - Otherwise proceed to the next step.
2. Unpack the crates.

Most crates are held together with metal clips. The first clip can be pried off using a screwdriver or similar object, and the remaining clips can be snapped off using one of the removed clips. It is suggested that the top panel be removed first, followed by the front, and one or more of the sides.

CAUTION: Keep one hand on each metal clip as it is pried off, because it can become airborne as a result of the clipping tension.

3. Locate and unpack the Power Rack.
4. Locate the tool bag in the Power Rack drawer.

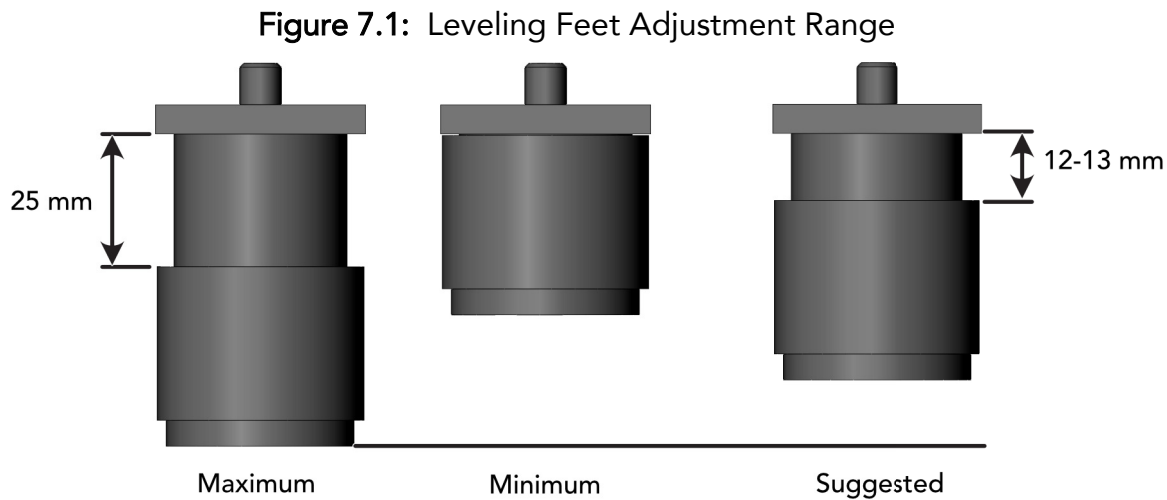
NOTE: The tool bag contains the tools that will be necessary for the rest of the installation.

7.2 Level the Robot Stand

The following is required for unpacking and leveling the robot stand:

- Two people to lift the robot stand
- A spirit level

The robot stand comes assembled. The only setup required is to level the stand using the adjustable feet.



1. Prepare the leveling feet:
 - a. Place the robot stand on its back.
 - b. Twist each of the four leveling feet by hand so that they are approximately in the middle of their range of adjustment. Each foot has a range of about 25 mm, so aim for 12-13 mm, as shown in [Figure: 7.1](#).

ATTENTION: This step reduces the chances of problems with leveling the Virtual Reality Stand in subsequent steps when a floor is uneven.

2. Place the robot stand upright on its feet and carry the stand to the desired location.
3. Use the level to determine if the robot stand is sloped from front-to-back or from side-to-side.
4. Adjust the feet until the robot stand is level in both directions.

7.3 Mount the Kinarm End-Point Robot

The following is required for unpacking and mounting the robot to the robot stand:

- Two people to lift each robot
- 5 mm Hex Key

Each Kinarm End-Point Robot comes fully assembled and packed in its own box. This section describes how to mount the robot to the robot stand that comes with the Kinarm End-Point Labs.

1. Remove the Kinarm End-Point Robot from the box.

- a. Remove the top foam insert from the box.

ATTENTION: The linkage is wrapped in plastic against the enclosure. Do not remove this wrapping until directed to remove it because it prevents the linkage from moving unexpectedly.

- b. With one person on each side of the robot, lift the robot out of the box.

ATTENTION: The Kinarm End-Point Robot will be damaged if it is lifted by any part of the linkage. Lift using only the built-in lifting handles as shown in [Figure: 7.2](#).

Figure 7.2: Lift the Kinarm Robot



2. Place the robot on the robot stand.

If installing a unilateral lab, the robot should be centred on the robot stand, and if installing a bilateral lab, the robots should be placed towards the sides of the robot (see [Figure: 7.3](#)). For the bilateral labs, the robots need to be attached to the stand with the links in the correct orientation. As in [Figure: 7.4](#), Link L3 should be closer to the centre of the stand than Link L1.

3. Attach each robot to the robot stand:

- a. Align the robot mounting holes with the robot stand mounting holes according to [Figure: 7.3](#).

The mounting tabs on the robot should be aligned with the robot stand horizontal supports. When the robot is in the correct position, the front of the robot will

overhang the robot stand, and the back of the robot will be flush with the back horizontal support of the robot stand.

- b. Insert and partially tighten two of the supplied 50 mm M6 socket head cap screws (with washers) into the front robot mounting holes (see [Figure: 7.4](#)).

ATTENTION: do not fully tighten the front screws.

- c. Insert and partially tighten the remaining two supplied 50 mm M6 socket head cap screws (with washers) into the back robot mounting holes.
- d. Tighten all four screws (front and back).

Figure 7.3: Robot Mounting Holes for Unilateral and Bilateral Labs

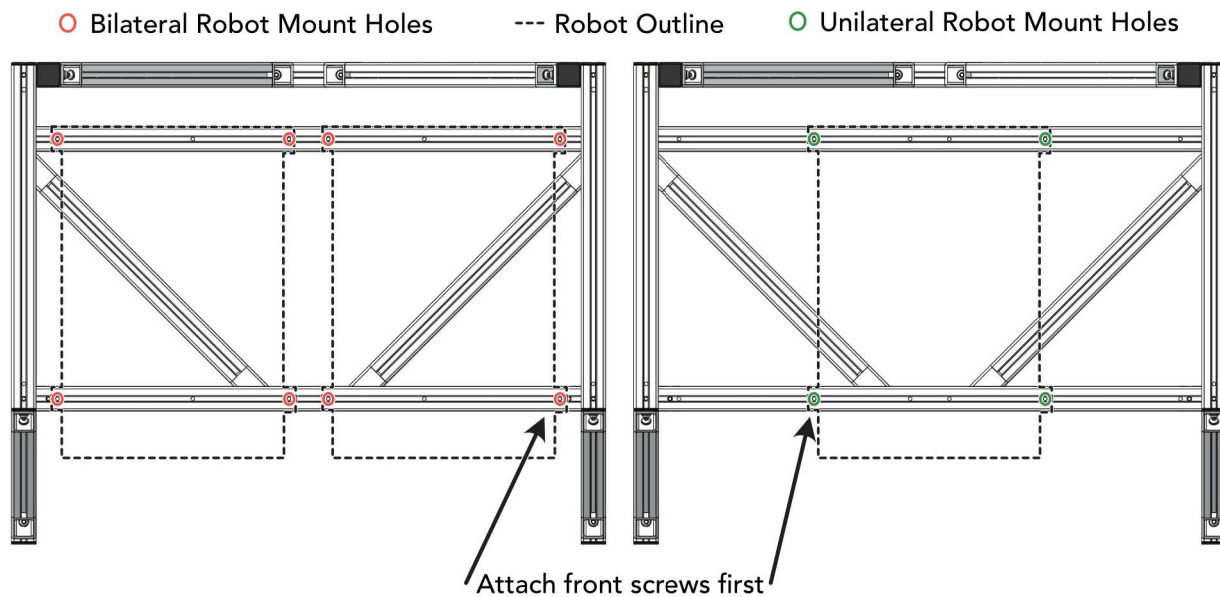
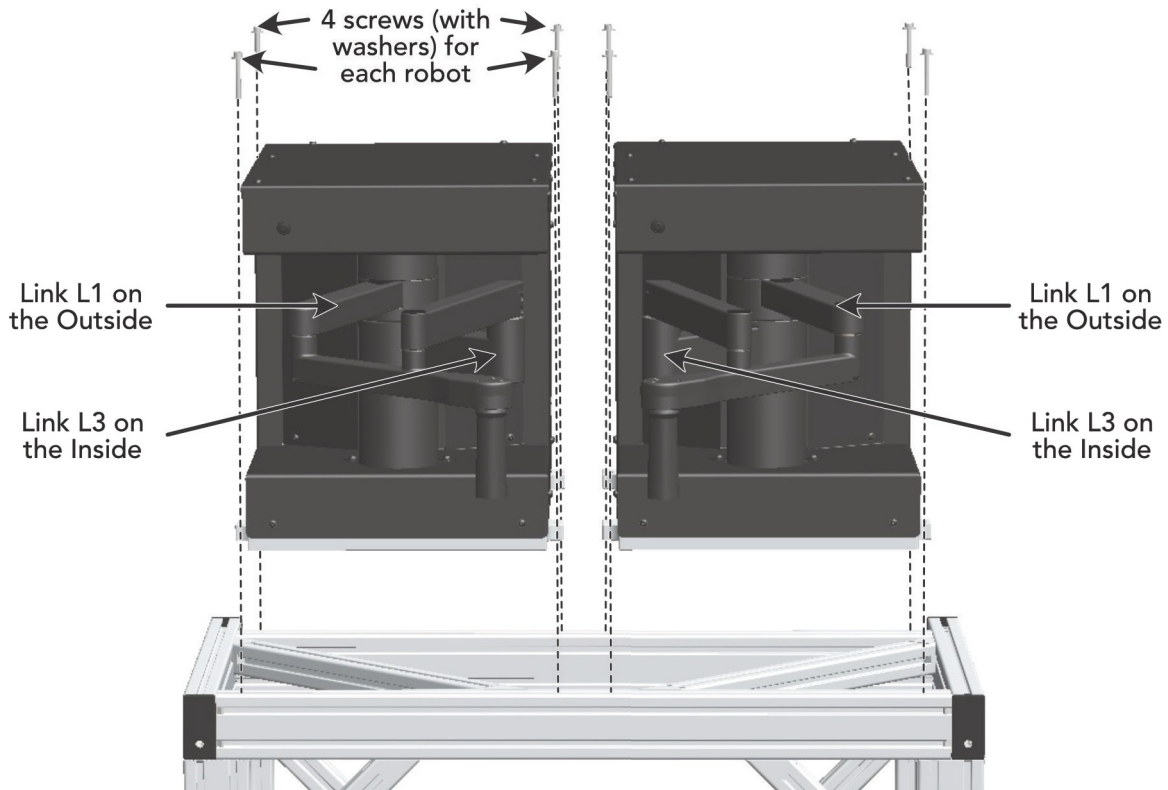


Figure 7.4: Mount the Kinarm Robot onto the Kinarm Stand

4. Remove the plastic wrap that prevents the linkage from moving.

7.4 Assemble the Table Top

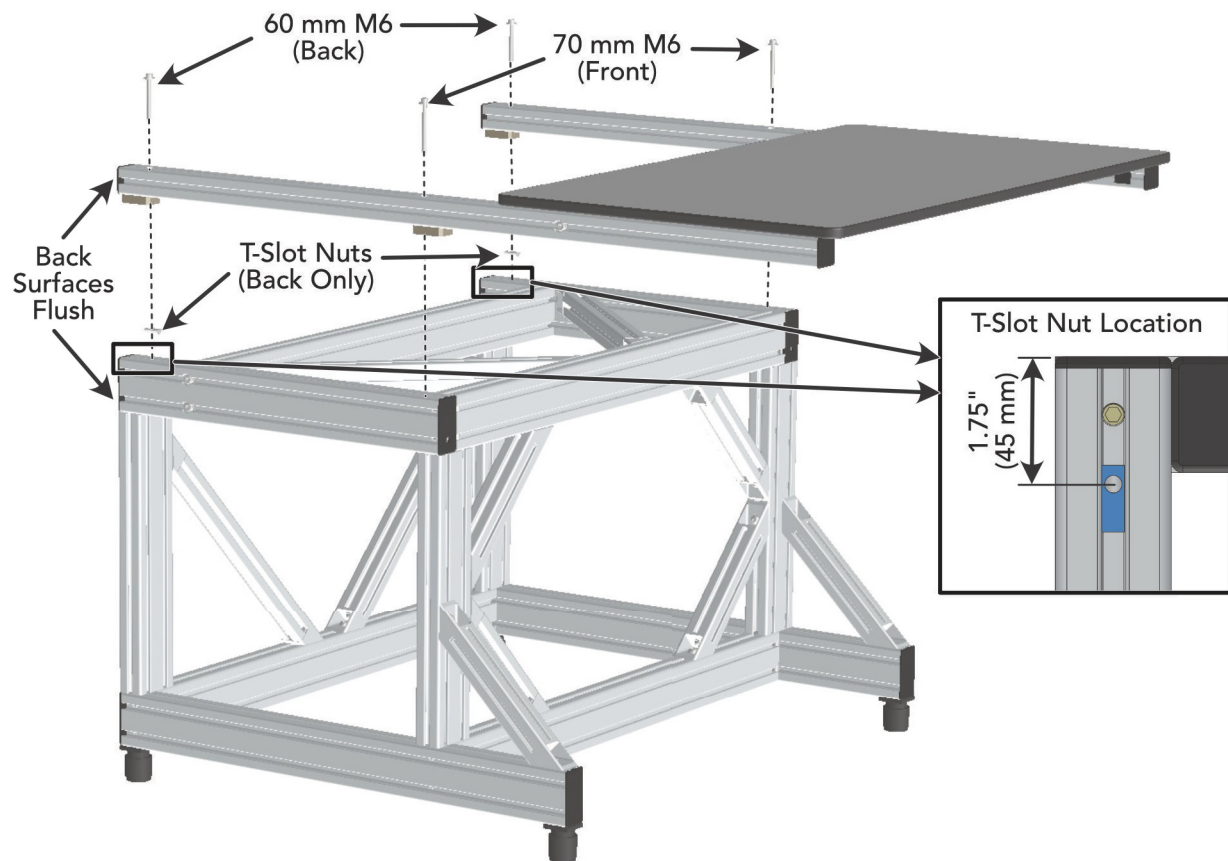
The following is required for unpacking and assembling the Table Top:

- Two people to lift the table top
- 5 mm Hex Key
- Tape measure or ruler

The table top mounts directly to the robot stand, and works with both the virtual reality display and the vertical display. The robot stand is shipped with the table top removed so it must be re-attached.

NOTE: The table top can be attached at any point during the assembly of the lab.

Figure 7.5: Install the table top onto the robot stand



1. Insert a supplied T-slot nut into each of the top sides of the robot stand approximately 1.75" (45 mm) from the back as illustrated by the inset of [Figure 7.5](#).
When inserted in the correct location, the hole in the nut will be aligned with the hole in the extrusion.
2. Position the table top such that the horizontal support ends are flush with the back of the robot stand.
3. Insert and partially tighten the two 60 mm M6 socket head cap screws (with washers) into the back holes of the table top supports until they are engaged in the t-slot nuts

ATTENTION: The front and back screws are different length. The 70 mm M6 screws are too long to fit in the back position, preventing the table top from tightening against to the robot stand.

ATTENTION: Do not fully tighten these screws yet.

4. Insert and partially tighten the two 70 mm M6 socket head cap screws (with washers) into the front holes of the table top supports.
5. Tighten all four screws (front and back).

7.5 Assemble the Virtual Reality Display

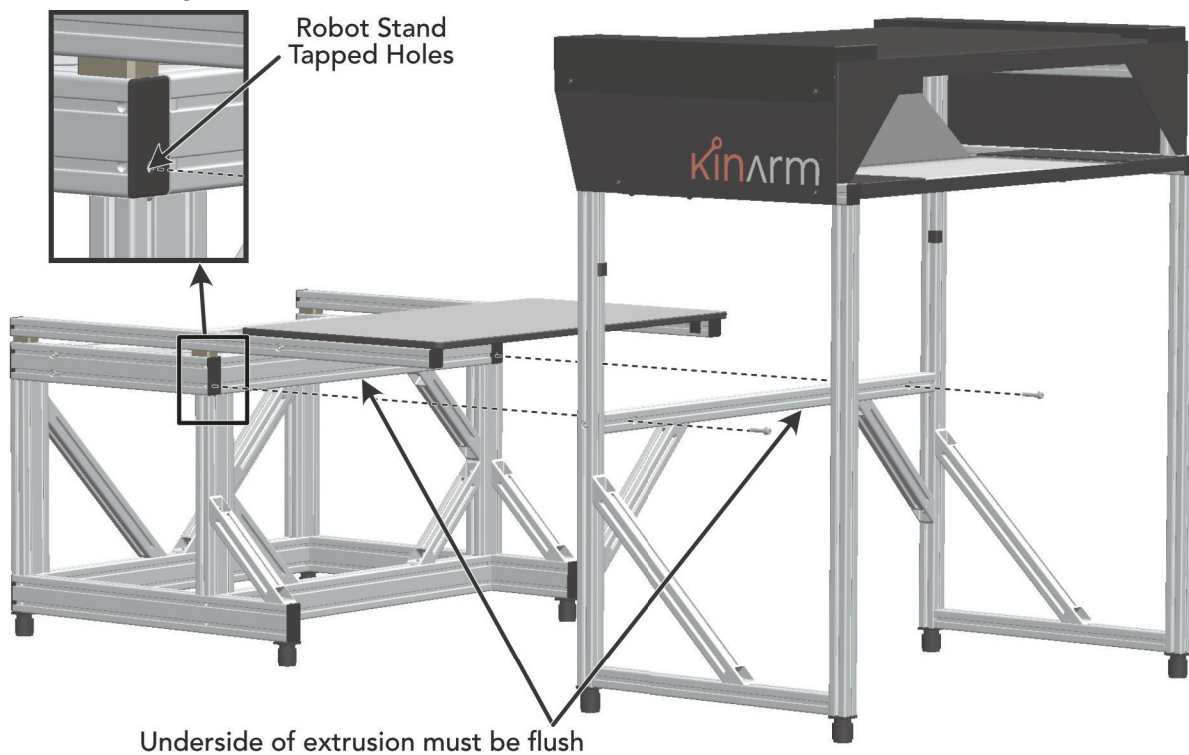
The following are required for unpacking and assembling the Virtual Reality Display:

- Two people to lift the Virtual Reality Display and subject display
- Spirit Level
- 4 mm and 6 mm Hex Keys
- Tape measure or ruler

The virtual reality display comes fully assembled, except for mounting the subject display and semi-transparent mirror. This section describes how to complete the assembly and attach the display to the robot stand.

1. Attach the virtual reality stand to the robot stand.

ATTENTION: Ensuring the virtual reality stand is aligned in height and level with the robot stand ensures that the projected images are correctly aligned with the robots.

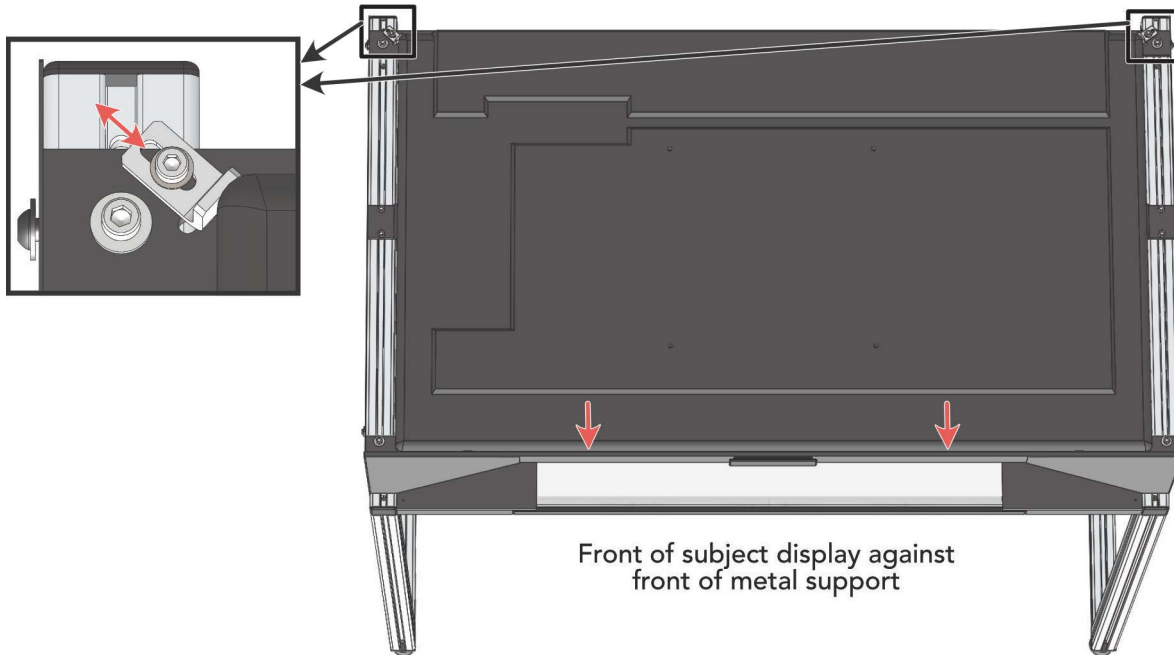
Figure 7.6: Attach the Virtual Reality Stand to the Robot Stand

- Lift the virtual reality stand into position.
The back of the virtual reality stand should be flush against the front of the robot stand.
- Adjust the height of both back feet of the virtual reality stand until the bottom of the horizontal extrusion of the virtual reality stand is flush with the bottom of the mating extrusion of the robot stand ([Figure: 7.6](#)).
- Adjust the height of the virtual reality stand's front leveling feet until the virtual reality stand is level from front-to-back.
Confirm the bottom of the horizontal extrusion at the back of the virtual reality stand is still flush with the bottom of the mating extrusion of the robot stand. If necessary, repeat the back and front feet height adjustments.
- Centre the virtual reality stand with respect to the robot stand (side-to-side).
If the height of the virtual reality stand is adjusted correctly (as per previous step), the screw holes in the virtual reality stand will align with the holes in the front end caps of the robot stand.
- Insert and partially tighten the two supplied 40 mm M8 socket head cap screws (with washers) through the horizontal extrusion of the virtual reality stand and into the two holes in the front of the robot stand.

ATTENTION: The screws should not be difficult to turn. If the screws do not easily thread into the robot stand, adjust the height and/or position of the virtual reality stand to more closely align the holes.

- f. Double-check that the bottom of the horizontal extrusion of the virtual reality stand is flush with the bottom of the mating extrusion of the robot stand ([Figure: 7.6](#)).
- g. Tighten the two 40 mm M8 screws.
2. If installing a Kinarm Gaze-Tracker, proceed with [7.9 Attach the Kinarm Gaze-Tracker](#) before continuing with the rest of this section.
3. Mount the subject display to the virtual reality stand.

Figure 7.7: Top View of the Installation of the Subject Display onto the Virtual Reality Stand



- a. Loosen the screws on the subject display clamps (see [Figure: 7.7](#)) and slide the clamps out of the way.
- b. Unpack the subject display from the box, and remove any protective plastic coverings.
- c. Place the subject display on top of the virtual reality stand so that the screen faces the floor and the top edge of the subject display is in contact with the front support ([Figure: 7.7](#)).
- d. Centre the subject display on the virtual reality stand (side to side).
Measure the distance from the outside edge of virtual reality stand to the edge of the subject display on each side. If necessary, shift the display sideways until the measurement is equal on both sides.
- e. Slide the subject display clamps against the subject display ([Figure: 7.7](#)) and tighten the screws.

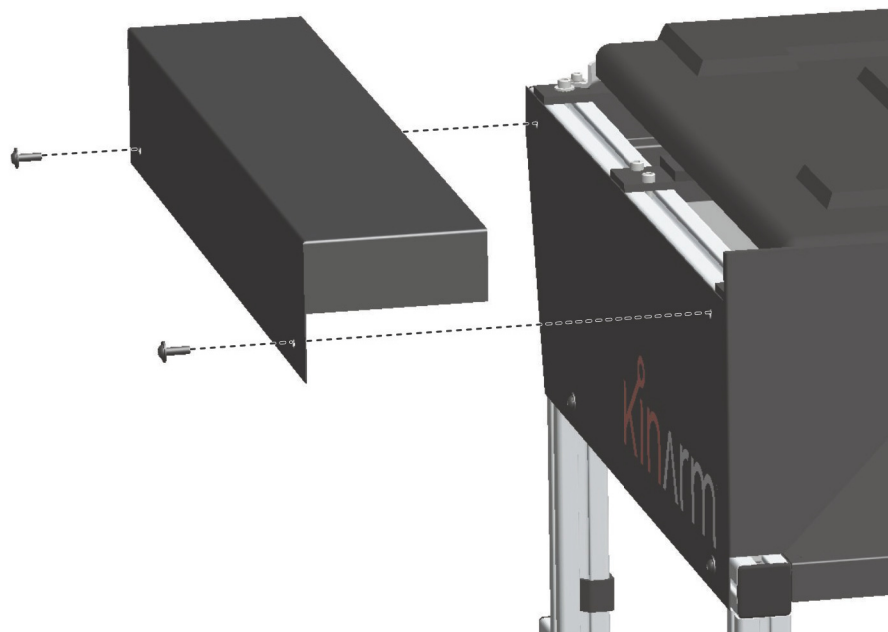
NOTE: The infrared sensor for the remote control of the subject display is typically mounted to the back left corner of the display which is the default installation location

for the display. The light blocking panels may partially cover the sensor, reducing the functional range of the remote control. Given that the only allowed function of the remote is the power switch, and that the display typically turns on and off automatically when the Dexterit-E Computer turns on, there is little need for the remote control. If better access for the remote control is desired, the infrared sensor may be unscrewed from the display and secured with a cable tie in a more accessible location.

4. Install the semi-transparent mirror.
 - a. Remove the side panels from the top of the mirror layer (see [Figure: 5.4](#)).
The side panels are attached by Velcro.
 - b. Unpack the semi-transparent mirror from the box, and place it onto the mirror layer with the reflective side facing upwards.
The reflective side is the side with the film applied to it. It should be possible to see and feel the edge of the film about 3 mm from the edge of the glass.
 - c. Reinstall the side panels on the mirror layer.
5. Attach the light blocking panels to the each side of the virtual reality stand.

NOTE: The screws that hold the light blocking panels in place are pre-installed, so they must first be removed

Figure 7.8: Installation of Light Blocking Panels.



- a. Remove the top two screws holding the side panels to the top piece of extrusion framing as shown in [Figure: 7.8](#).

- b. Position the light blocking panel in place, lining up its holes with the two holes where the screws were just removed.
The left and right side panels are not quite symmetrical and may be difficult to differentiate. Typically they are labeled when packaged for initial installation. However if the labels are not present, pick one panel and try to align the holes on one side. If the holes at the front will not line up, then install the panel on the other side of the virtual reality stand.
- c. Re-install the two screws.

7.6 Assemble the Vertical Display

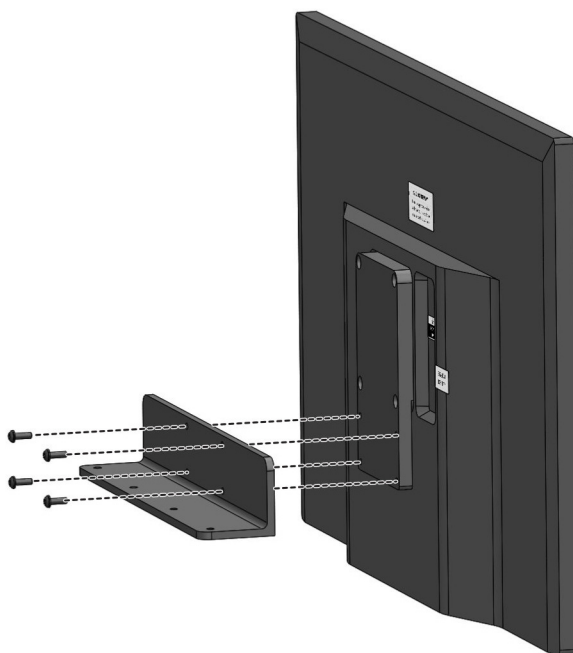
The following is required for unpacking and mounting the Vertical Display:

- Two people to lift the subject display.
- 3 mm Hex Key

The vertical display mounts directly to the top of the robots in either unilateral or bilateral configurations. This section describes how to attach the display to the robots.

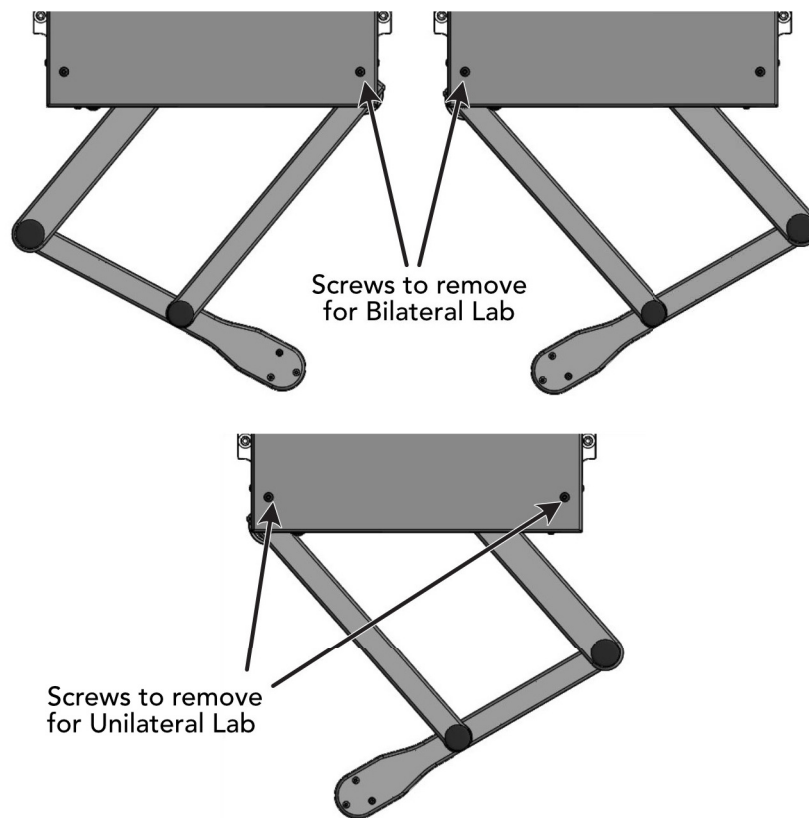
1. Attach the mounting bracket to the subject display:
 - a. Unpack the subject display from the box, and remove any protective plastic coverings.
NOTE: The vertical display typically comes with a mounting plate pre-attached to the VESA mounts on the display.
 - b. Attach the mounting bracket to the VESA display mounts using the four supplied 16 mm M6 button head cap screws (with washers) as shown in [Figure: 7.9](#).

Figure 7.9: Attach the Mounting Brackets to the Vertical Display



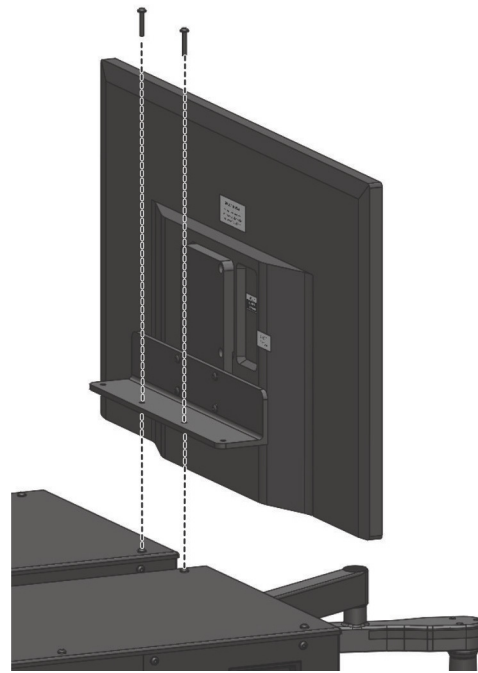
2. Mount the subject display to the robot(s):
 - a. Remove the following screws from each robot (see [Figure: 7.10](#)):
 - Unilateral lab - Remove both screws located on the top front of the robot.
 - Bilateral lab - Remove the left top front screw from the right robot and the right top front screw from the left robot.

NOTE: These screws are no longer needed, however, they should be kept in case of a future upgrade to a virtual reality stand.

Figure 7.10: Remove Screws from Top of Robots

- b. Hold the subject display in place so that the mounting bracket rests upon the robots. The subject display should be centred across the robot(s), and the holes in the mounting bracket should line up with the holes exposed by the two removed screws (see [Figure: 7.11](#)). There are two pairs of holes in the mounting bracket. The inner pair is for a bilateral system, and the outer pair is for a unilateral system.

Figure 7.11: Mount the Vertical Display onto the Robots



- c. Insert the two supplied 30 mm M5 button head cap screws (with washers), into the available holes.
- d. Tighten the two screws.

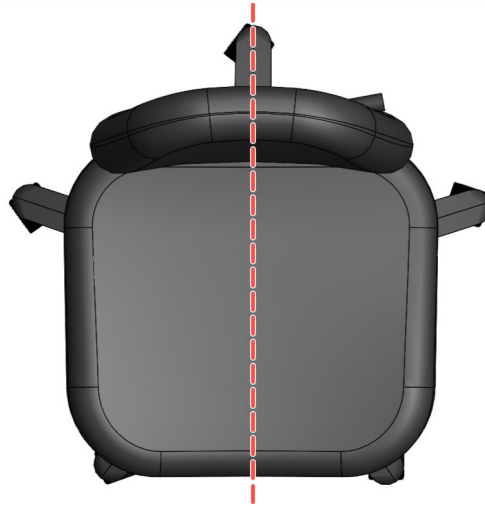
7.7 Assemble the Chair

The Standard and Deluxe Chairs are largely pre-assembled. No tools are required for the following assembly instructions.

7.7.1 Assemble the Standard Chair

The Standard Chair typically comes in four pieces in a box: base (including 5 casters with brakes), cylinder, seat and backrest.

1. Place the chair base (with 5 casters) on the floor and apply the brakes.
2. Insert the cylinder (widest end) into the central hole of the chair base. It is not necessary to press hard at this stage.
3. Lightly place the seat onto the top of the cylinder (there is a single hole in the bottom of the seat).
4. Turn the seat until it is facing the desired direction relative to the casters. Typically, the seat is oriented symmetrically relative to the base such that there is a single caster protruding from the back of the seat, as shown in [Figure: 7.12](#).

Figure 7.12: Standard Chair Orientation

5. Press (or sit) on the seat to ensure that the cylinder is completely press-fit into the seat and base.
6. Install the backrest.
If there is a plastic plug pressed into the backrest stem, remove it before inserting the backrest. Reinstall the plug after the backrest is attached to the chair.

7.7.2 Assemble the Deluxe Chair

The Deluxe Chair typically comes in three pieces: base (including seat, casters, belt and wired control), backrest (including harness), and hand brake lever.

1. Place the base on the floor and engage the brakes using the foot pedal.
2. Install the hand lever into the brake mechanism. The mounting hole for the lever is located near the rear-right caster. The lever slots into the hole of the same shape.

NOTE: The Deluxe Chair is initially set up with the hand lever mounted on the right side of the chair. The hand lever can be installed on the left side of the chair if preferred. Contact Kinarm Support for assistance.

3. Install the backrest, ensuring that it is inserted far enough for the push-pin to click into place.
4. Attach the wired remote to the mount bracket on the back of the chair.
The remote attaches magnetically, so it can be attached by bringing it in close proximity to the metal mount bracket.

NOTE: The chair will enter a stand-by mode after being inactive for 7 days (which is common during shipping). To wake it up, please see [11.7 Deluxe Chair Seat Does Not Move Up or Down](#).

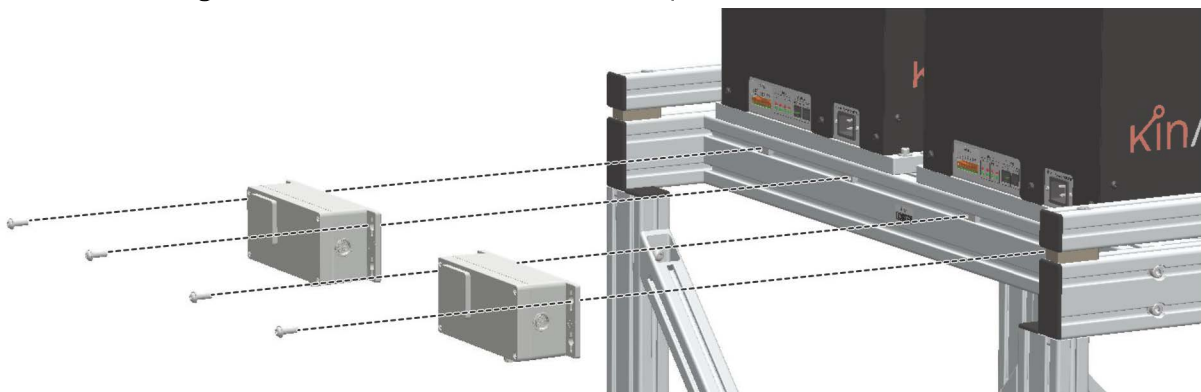
7.8 Attach the Force/Torque Sensor Interface Unit

The following is required for unpacking and mounting the force/torque sensor interface unit:

- 4 mm Hex Key

Each force/torque sensor requires a dedicated interface, which serves to communicate between the sensor itself and the computers on the Dexter-E LAN. Additionally, each interface unit is configured to work with a specific robot. Match the label on the interface unit with the label on the robot. Although these interface units can be mounted anywhere, it is suggested that they be mounted to the back support of the robot stand, just underneath the associated robot. They can be mounted to either the front or back surface of this back support; the images in this guide show mounting to the back surface, but mounting on front surface may allow easier access/visibility for trouble-shooting. Alternatively, the interface boxes can be attached to the side of the robot stand, but they need to be placed as close to the back of the stand to ensure that the sensor cables do not interfere with movement of the linkage.

Figure 7.13: Attach the Force-Torque Sensor Interface Unit



1. Insert two T-slot nuts for each interface box into the extrusion and align them with the mounting holes of the interface unit.([Figure: 7.13](#))
2. Insert and tighten two supplied 20 mm M6 button head cap screws (with washers) through each interface box and into the T-slot nuts.

7.9 Attach the Kinarm Gaze-Tracker

The following is required for unpacking and mounting the Kinarm Gaze-Tracker:

- 5/32" Imperial Hex Key
- 3/16" Imperial Hex Key

The Kinarm Gaze-Tracker must be mounted within the Virtual Reality Display. However, if the Kinarm Gaze-Tracker is being installed at the same time as the virtual reality display, then the first step can be skipped.

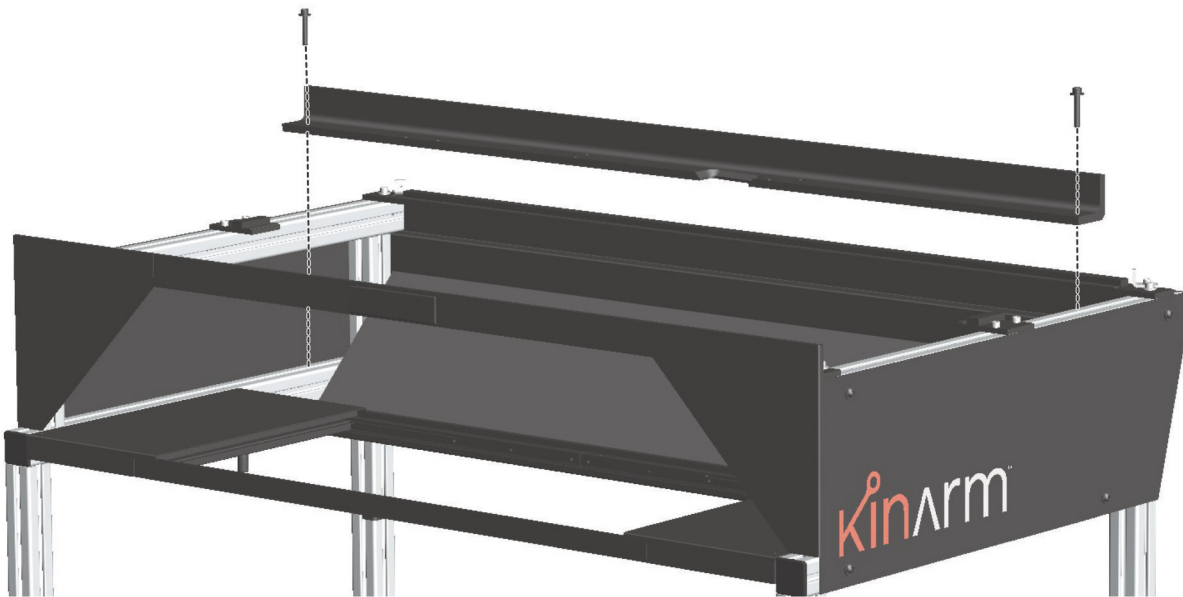
ATTENTION: The following instructions apply only when the Kinarm Gaze-Tracker is purchased at the same time as the Virtual Reality Display. If an existing Virtual Reality Display is being

upgraded to include a Kinarm Gaze-Tracker, the instructions below are not sufficient. Contact Kinarm Support for instructions on how to upgrade.

1. Remove the light-blocking panels, subject display and semi-transparent mirror from the Virtual Reality Display if they are already installed. There is typically no need to loosen the subject display clamps.
[7.5 Assemble the Virtual Reality Display](#) can be used a reference as needed.
2. Attach the Kinarm Gaze-Tracker main support angle to the Virtual Reality Display (see [Figure: 7.14](#)).
 - a. Insert two black screws (1/4-20 x 1.5" SHCS, and associated washers). The screws insert directly into the aluminum extrusion framing.

NOTE: The Virtual Reality Display is typically shipped with this main support angle already pre-installed.

Figure 7.14: Attach the Main Support Angle to the Virtual Reality Display

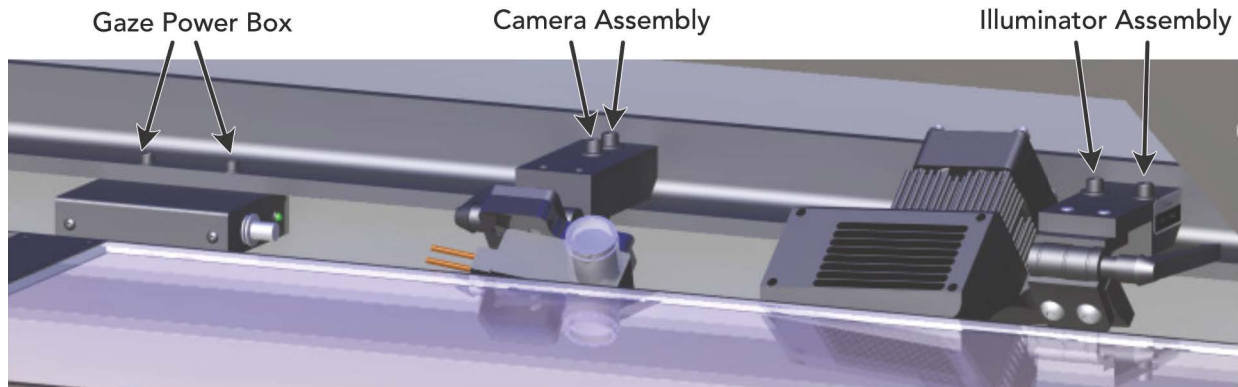


3. Attach the gaze components to the main support angle (see [Figure: 7.15](#)).
 - a. Using the two SHCS #10-32 x 0.625" screws (and associated washers), attach the gaze power box underneath the main support angle (the screws attach from the top).
 - b. Using two of the black SHCS 1/4-20 x 1.5" screws (and associated washers), attach the camera assembly on top of the main support (the screws attach from the top).
 - c. Using the remaining two black SHCS 1/4-20 x 1.5" screws (and associated washers), attach the illuminator assembly on top of the main support (the screws attach from the top).

- d. Set the angles of both the camera head and illuminator approximately 45 degrees using the locking hinges.

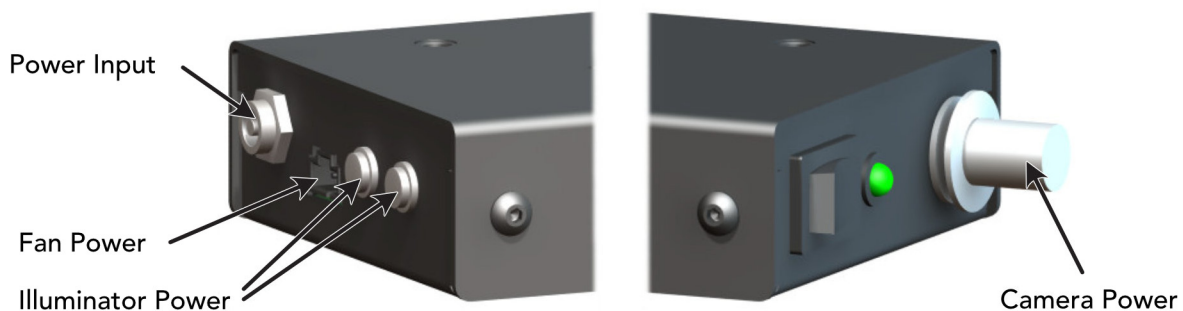
NOTE: The camera and illuminator assembly angles will need to be adjusted after the rest of the system is connected and functional, see [7.11 Adjust the Kinarm Gaze-Tracker](#)

Figure 7.15: Attach the Kinarm Gaze-Tracker Components



4. Connect the cables to the Gaze Power Box (see [Figure: 7.16](#)):
The cables should be routed neatly behind the components so that they do not block or interfere with adjustment of the components.

Figure 7.16: Gaze Power Box Connections



- a. Connect the two power cables (1/8" stereo jacks) from the illuminator.

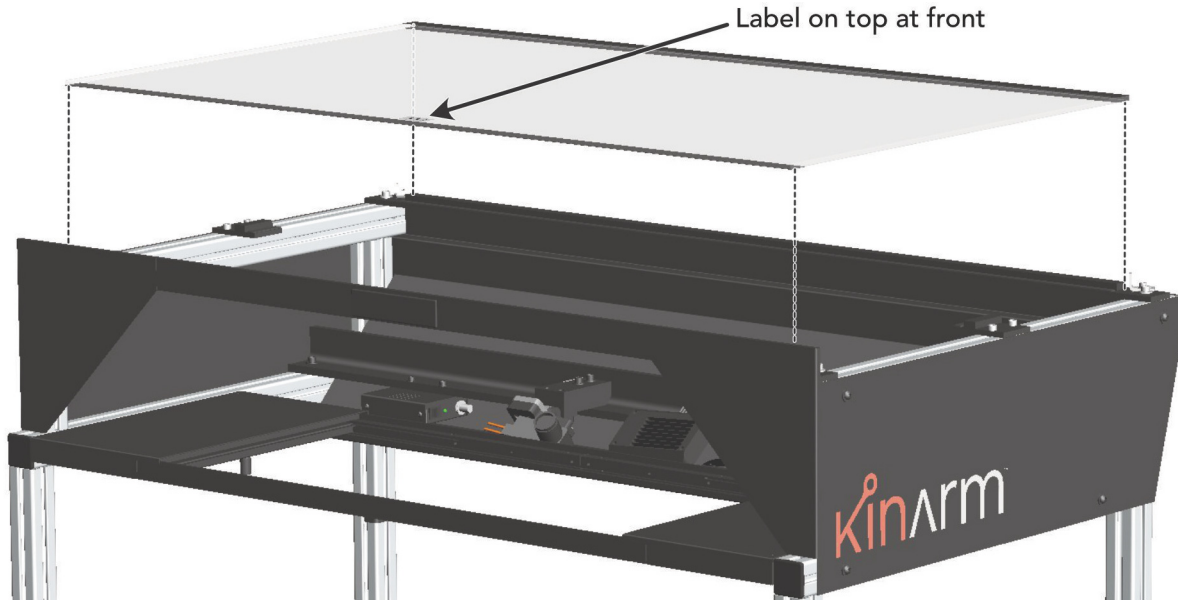
NOTE: Either cable can be attached to either jack.

- b. Connect the fan power cable.
- c. Connect the camera power cable.

NOTE: The flange on the gaze power box connector needs to be depressed in order to remove this cable.

5. Install the Hot Mirror, as shown in [Figure: 7.17](#).
The hot mirror will have a label on the top surface at the front. Be sure to install the hot mirror with the label facing upwards and at the front.

Figure 7.17: Install the Hot-Mirror

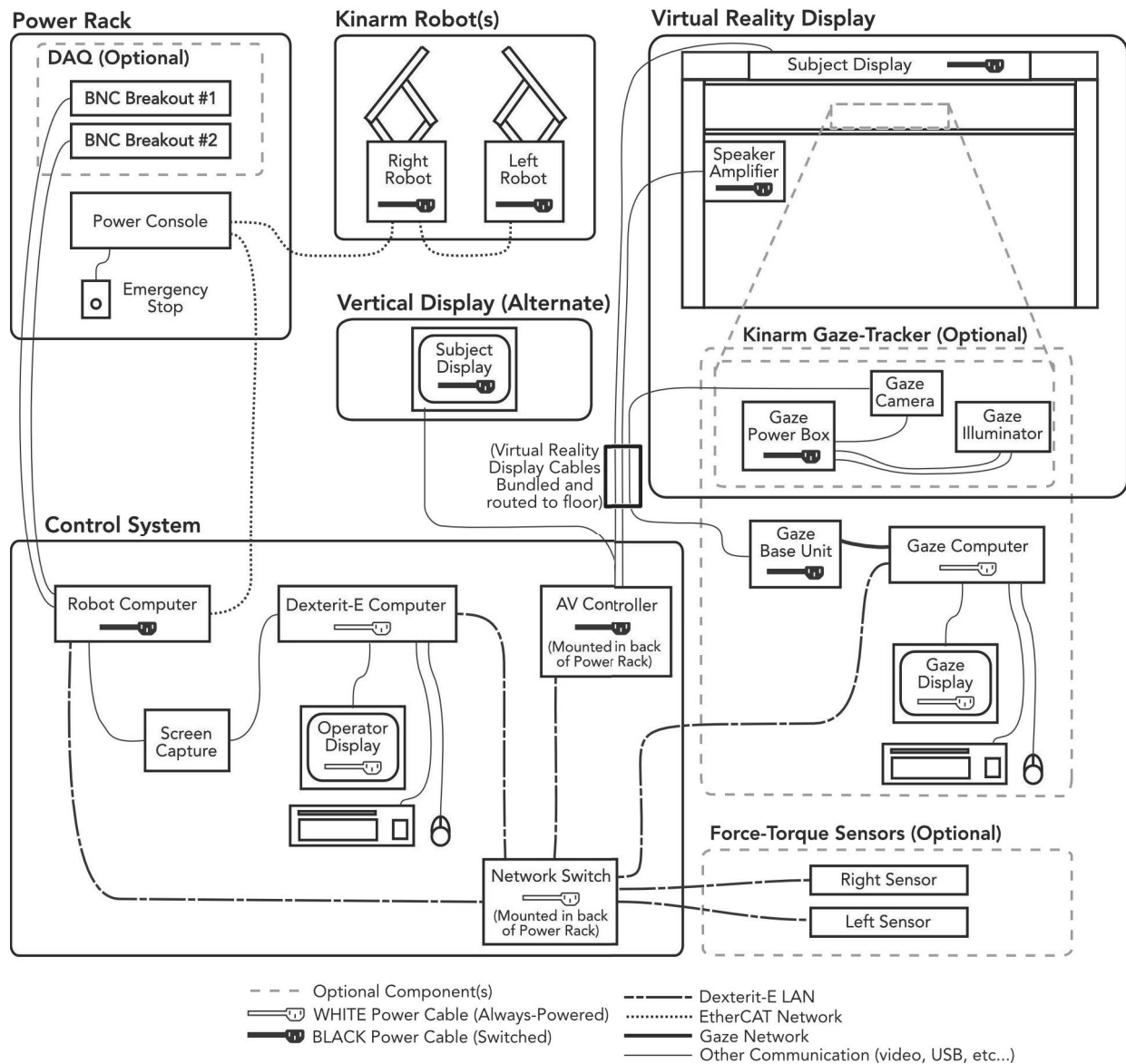


6. Install the subject display and continue the installation of the Virtual Reality Display as per [7.5 Assemble the Virtual Reality Display](#).

7.10 Connect the Lab Components

To complete the lab setup, all of the individual components must be connected together for communication between the components. An overall schematic of how these components need to be connected is shown in [Figure: 7.18](#).

Figure 7.18: Kinarm Lab Schematic



The Kinarm Lab is configured at the factory to run off the appropriate voltage for your installation. Refer to [6.3 Electrical](#) for the electrical requirements for the Kinarm Lab.

WARNING: To avoid the risk of electric shock, this equipment must be connected to a supply mains with protective earth.

Power to all the components is provided by the Power Console, which is mounted in the Power Rack. Refer to section [8.2 Powering the Kinarm Lab Components](#) to determine which ports should be used to power the various components of the Kinarm Lab. The Power Console is also

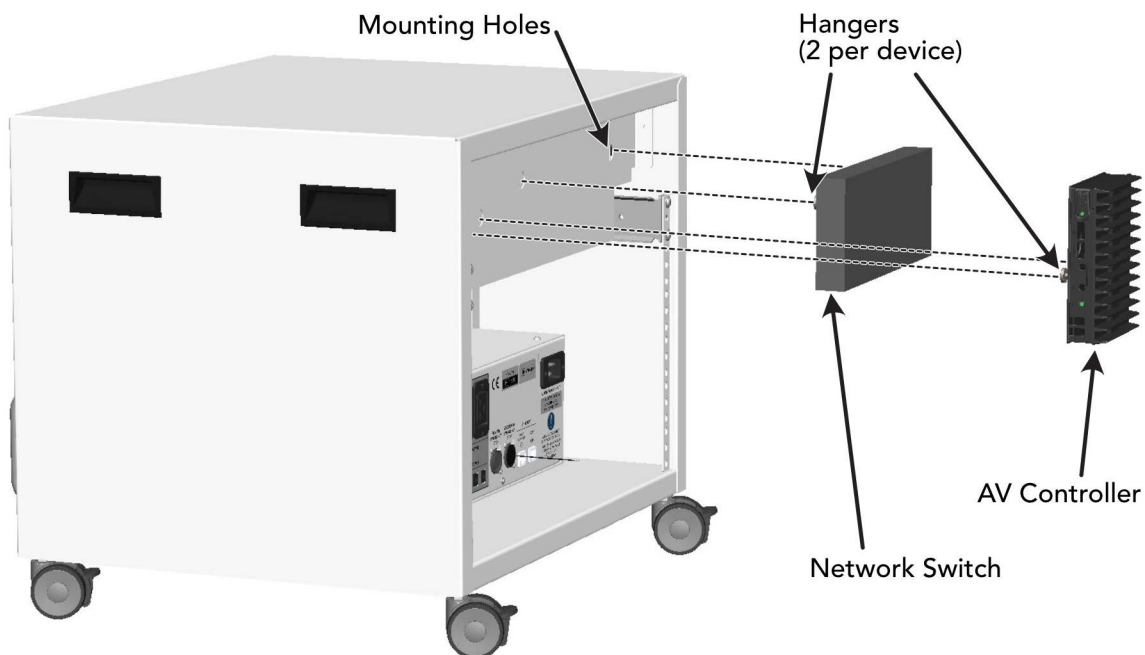
the hub for communication cables including the EtherCAT network (motion control) and the Emergency Stop Button.

WARNING: To avoid the risk of electric shock, do not connect anything that is not part of the Kinarm Lab to the Power Console.

1. Unpack and position the remaining components.
 - a. Unpack the components from their shipping boxes.
Cables that connect to specific components are often shipped in that component's box.
 - b. Unpack any cables that were shipped in the drawer of the Power Rack.
 - c. Position the components in the desired location (see [6.7 Kinarm Lab Placement](#) for guidance).
The components do not need to be placed precisely, but placing them roughly where they will end up will simplify the routing of cables.

2. Prepare the Power Rack.

Figure 7.19: Mounting the AV Controller and Network Switch



- a. Set up the network switch:
 - i. Plug the power adapter into the network switch.
 - ii. Hang the network switch on the rear panel of the Power Rack using the pre-installed hangers such that the network ports are facing downwards (see [Figure: 7.19](#)).
 - iii. Plug the power adapter into the Power Console (**WHITE**, 1 m).

- iv. Tuck the power adapter and excess power cable to the side of the Power Console.
- b. Set up the AV Controller:
 - i. Plug the one end of the power cable (**AV CONTROLLER POWER**) into the AV Controller and the other end into the Power Console.
 - ii. Plug one end of the short (~0.6 m) yellow network cable (**DEX LAN**) into the AV Controller and the other end into any one of the ports on the network switch.
 - iii. Hang the AV Controller on the rear panel of the Power Rack using the pre-installed hangers such that the power cable is facing towards the network switch (see [Figure: 7.19](#)).
- c. Plug the hand-held emergency stop into the **PRIMARY EMERGENCY STOP** port on the Power Console.
3. Connect the Kinarm Robot cables.
 - a. For each Kinarm Robot, plug the power cable (**BLACK, POWER CONSOLE TO ROBOT POWER**) from the Power Console (any available port) to the robot.
 - b. Plug one black network cable (**ETHERCAT**) from the **ROBOT** port of the Power Console to the **ETHERCAT IN** port on the right (or unilateral) robot.
 - c. If applicable, plug another black network cable (**ETHERCAT**) from the **ETHERCAT OUT** port on the right robot to the **ETHERCAT IN** port on the left robot
- ATTENTION:** The order of the robot connections for bilateral labs is important. The left robot must be connected at the end of the EtherCAT network.
4. Connect the Force/Torque Sensor cables (as applicable).
 - a. For each sensor, attach the sensor cable (not labeled, coming from the robot linkage) to the **TRANSDUCER** port on the matching Force/Torque Sensor Interface Unit. Ensure that the connector is fully threaded into the port and that the Interface Unit matched the sensor as noted in [7.8 Attach the Force/Torque Sensor Interface Unit](#).
 - b. For each sensor, plug the threaded end of a yellow network cable (**DEX LAN**) into the **ETHERNET** port on the Force/Torque Sensor Interface Unit, and other end into any one of the ports on the network switch.

ATTENTION: It is important that all connectors are fully threaded into the ports on the Force/Torque Sensor Interface Unit.

NOTE: Power is provided to the Force Torque Sensors via PoE (Power Over Ethernet) through the yellow network cables.

5. Connect the Virtual Reality Display cables (if applicable).
 - a. If applicable, set up the 120 Hz subject display power adapter on the Virtual Reality Display:
 - i. Attach the power supply to the Virtual Reality Display using the pre-installed Velcro strips.
 - ii. Plug the power supply into the display.
 - iii. Turn the power switch on the back of the display ON.
 - b. Plug one end of the following cables into the Virtual Reality Display components:
 - Power cable (**BLACK**) into the subject display (standard display) or into power adapter (120 Hz display).
 - Video cable (**SUBJECT DISPLAY**) into the port on subject display with matching label.
 - Power cable (**SPEAKER: AMPLIFIER POWER INPUT**) into the amplifier power input port.
 - Audio cable (**AUDIO OUT TO AMP IN**) into the amplifier audio input port.
 - If applicable - Gaze Power Box power adapter into the Gaze Power Box (see [Figure: 7.16](#)), and Power cable (**BLACK**) into the gaze power box power adapter. Power adapter should be plugged into the Gaze Power Box as per [Figure: 7.16](#). Route the output cable from the back of the Virtual Reality Display, around the left side of the shadow blocker panel (i.e. left from the perspective of a subject), to the Gaze Power Box..
 - If applicable - Fibre-optic cable into Gaze camera. Route into the Virtual Reality Display along with the gaze power box power cable.

NOTE: The Gaze Power Box and Fibre-optic cables should be routed from the back of the Virtual Reality Display, around the left side of the shadow blocker panel (i.e. left from the perspective of a subject), to the Gaze Power Box and Camera respectively.
 - c. Secure the cables to one side of the Virtual Reality Display.

The cables should be routed towards the rear on the side that is closest to the Power Rack and computers and down to the floor.

 - i. As needed, tuck cables into the routing panels on the back of the shadow blocker panel to prevent cable from drooping onto the robots.
 - ii. Gather all cables into a single bundle on the desired side.
 - iii. Secure the bundle of cables to the three cable tie mounts on the rear vertical extrusion using cable ties, ensuring that the cables are straight and remain behind the extrusion.
 - d. Guide the bundle of cables towards the Power Rack and computers.
 - e. Plug other end of the following cables into the Power Console:
 - Power cable(s) (**BLACK**).

- Power cable (**SPEAKER: POWER CONSOLE POWER OUTPUT**).
 - f. Plug other end of the following cables into the AV Controller:
 - Video cable (**SUBJECT DISPLAY**) with adapter into the mini-DisplayPort port.
 - If applicable - 120 Hz subject display video adapter USB cable into any USB port.
 - Audio cable (**AUDIO OUT TO AMP IN**) into the audio output port.
 - g. If applicable, plug the other end of the following cables into the Gaze Base Unit:
 - Fibre-optic cable.
6. Connect the Vertical Display cables (if applicable).
- a. Plug one end of the following cables into the Vertical Display components:
 - Power cable (**BLACK**) into the subject display.
 - Video cable (**SUBJECT DISPLAY**) into the port on subject display with matching label.
 - b. Guide the bundle of cables away from the robots, down to the floor and towards the Power Rack and computers.
 - c. Plug other end of the following cables into the Power Console:
 - Power cable(s) (**BLACK**).
 - d. Plug other end of the following cables into the AV Controller:
 - Video cable (**SUBJECT DISPLAY**) with adapter into the mini-DisplayPort port.
7. Connect the remaining control system cables:
- Power cables (**BLACK, WHITE**) from the components into the Power Console.
 - Video cable (**OPERATOR DISPLAY**) into one of the ports labeled **KINARM DISPLAYS** on the Dexterit-E Computer.
 - Screen Capture Device:
 - Video cable (**ROBOT COMPUTER TO SCREEN CAPTURE**) and adapter from the port labeled **KINARM DISPLAYS** on the Robot Computer to the **INPUT** port on the screen capture device.
 - USB cable (**SCREEN CAPTURE TO DEXTERIT-E COMPUTER**) from the USB port on the screen capture device to any USB port on rear of the Dexterit-E Computer.
 - Mouse and keyboard USB cables into any USB port on the Dexterit-E Computer via extension cables as needed.
 - Yellow network cables (**DEX LAN**) from the network port labeled **DEX LAN** on each of the Dexterit-E and Robot Computers to any ports on the network switch
 - Black network Cable (**ETHERCAT**) from the network port labeled **ETHERCAT** on the Robot Computer to the network port labeled **ROBOT COMPUTER** on the Power Console.

- If applicable - DAQ cable(s) (**BNC BREAKOUT TO ROBOT COMPUTER**) from the port(s) on the PCI card in the Robot Computer to the BNC breakout panel(s) on the Power Rack.

NOTE: If only one BNC 2090A breakout board is installed, ensure that the DAQ cable is plugged into the port labeled **CONNECTOR 0 CHANNEL 0-15** on the PCI card.

8. If applicable, connect the remaining Kinarm Gaze-Tracker Computer system cables:
 - Power cables (**BLACK, WHITE**) from the components into the Power Console.
 - Video cable (**GAZE DISPLAY**) into the video port on the Gaze-Tracker Computer.
 - Mouse and keyboard USB cables into any USB port on the Gaze-Tracker Computer via extension cables as needed.
 - Yellow network cable (**DEX LAN**) from the network port labeled **DEX LAN** on the Gaze-Tracker Computer to any port on the network switch.
 - Black network cable (**GAZE**) from the network port labeled **GAZE** on the Gaze-Tracker Computer to the network port labeled **GAZE** on the Gaze Base Unit.
 - Gaze Base Unit power adapter (barrel) into the Gaze Base Unit.
9. Optional connections (not provided with Kinarm Lab):
 - a. Internet. Connect a network cable to the port labeled **INTERNET** on the Dexterit-E Computer.

ATTENTION: Do not connect the Internet to the DEX LAN network switch.
 - b. External USB Storage. Connect the USB cable to any USB port on the Dexterit-E Computer.

NOTE: All USB ports on the back of the Dexterit-E Computer are USB 3.0 ports (i.e. fast), while USB ports on the front are USB 2.0 (i.e. slow).
 - c. User DIO. A custom cable is required to connect to the User DIO ports on the Kinarm Chair. See [9.2 Robot User Digital Inputs and Outputs \(DIO\)](#) for details.
10. Plug the Power Console Power cable into the Power Console mains input, and then into the wall outlet.

7.11 Adjust the Kinarm Gaze-Tracker

The following may be required for adjusting the assemblies:

- 3/16" imperial hex key
- A piece of plain white paper
- Tape

When everything is properly connected, the camera and illuminator assemblies need to be adjusted to the correct angles.

ATTENTION: The Gaze-Tracker Computer comes preset with all the required settings. Under normal use, there is no need to change any settings because Dexter-E automatically specifies the correct settings. Any changes may prevent the gaze tracking from working correctly. If there are any problems with the Gaze-Tracker Computer settings, please contact Kinarm Support for assistance.

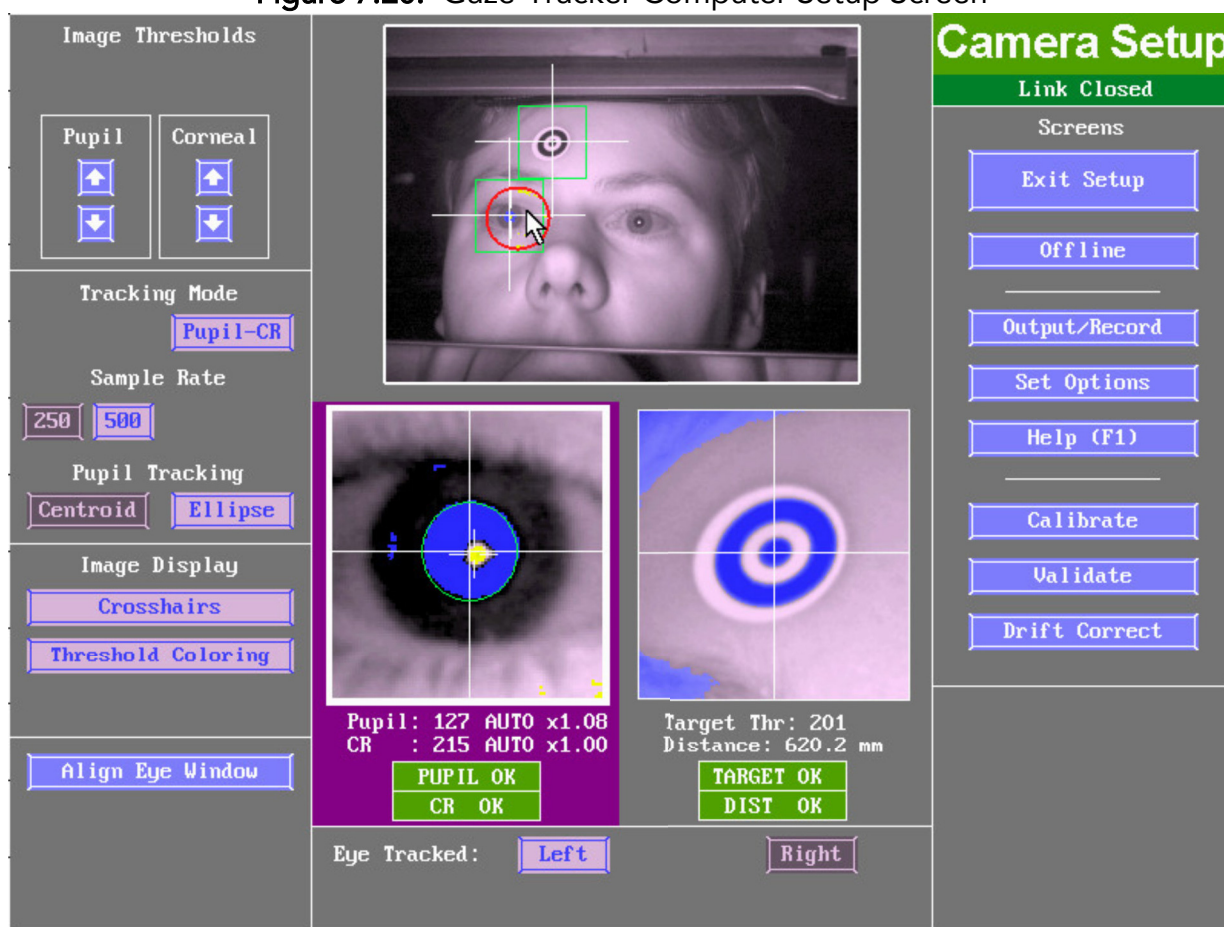
1. Turn on the Gaze-Tracker Computer.

The computer should automatically run the EyeLink Software. If given the option, select the EYELINK OS.

NOTE: If the Gaze-Tracker Computer reports a series of errors when starting up, cycle the power on the camera base box and try again.

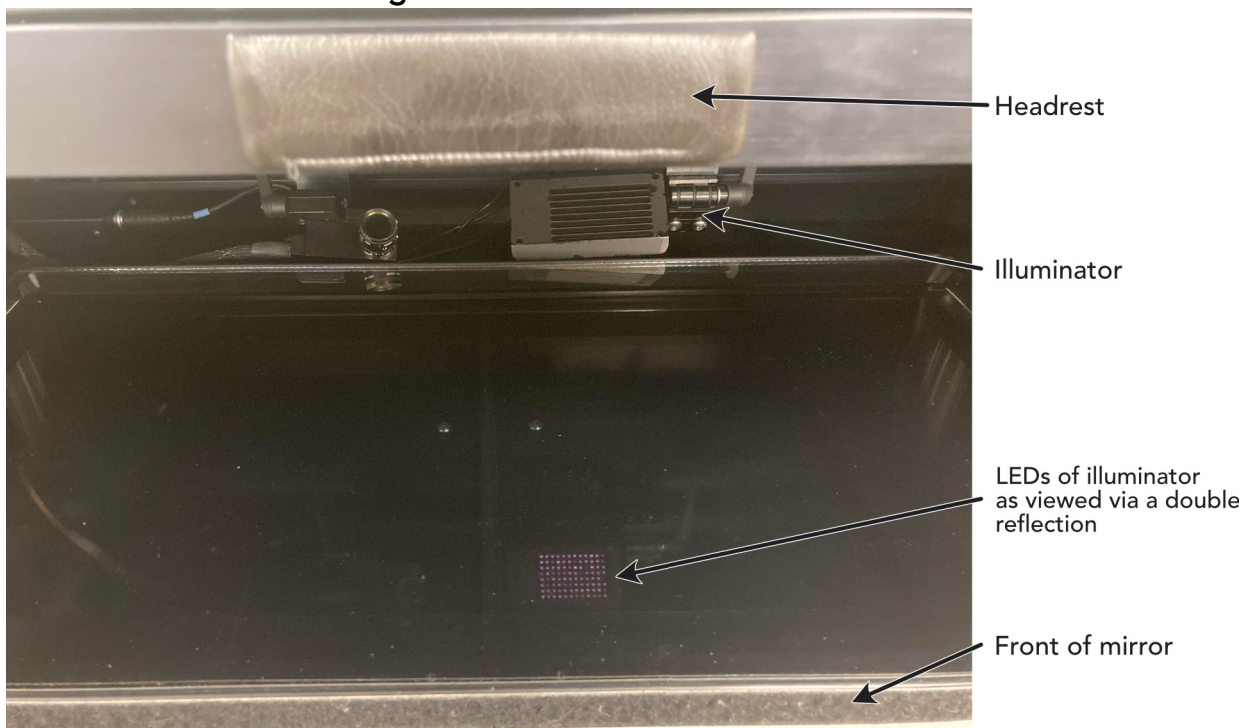
2. Navigate to the Setup Screen, as shown in [Figure: 7.20](#).

Figure 7.20: Gaze-Tracker Computer Setup Screen



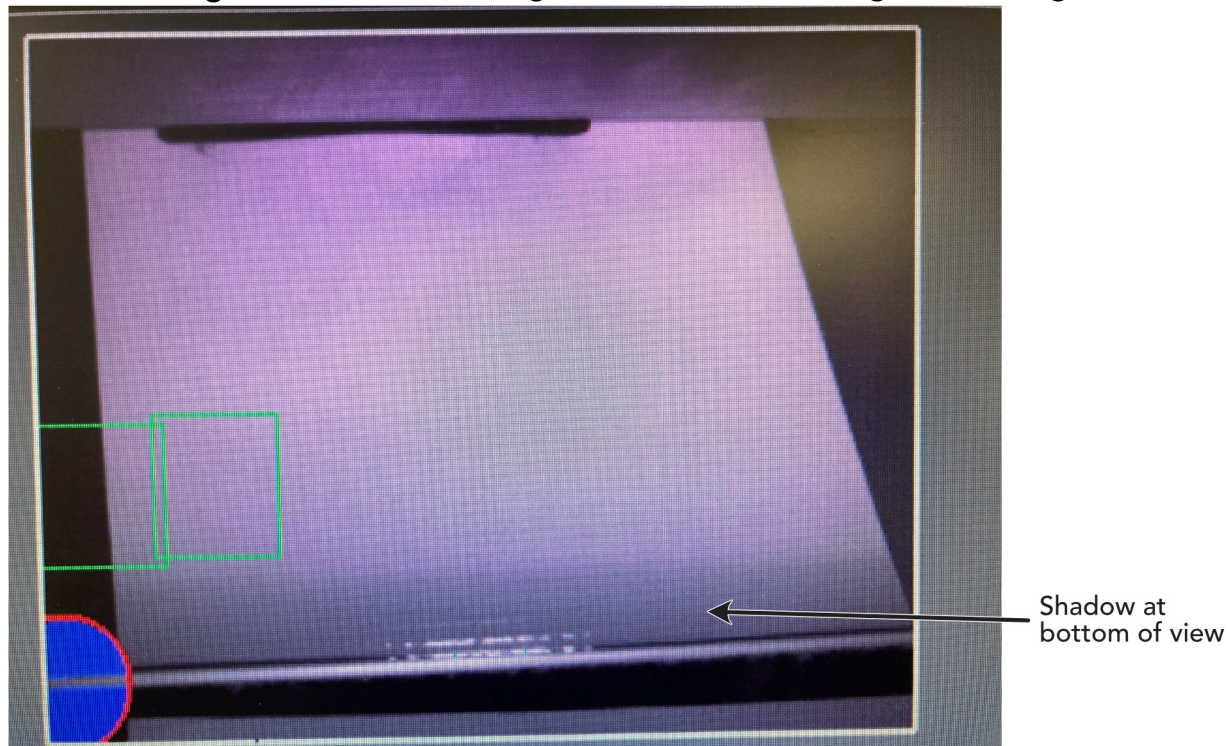
3. Adjust the camera assembly angle:
 - a. Using the locking hinge, adjust the angle of the camera head so that the image of the subject's head is right-side up (not upside down) and is centred vertically on the Virtual Reality Display (i.e. the top and bottom of the camera's view should show roughly equal amounts of the virtual reality display framing, as shown in [Figure: 7.20](#)).
 - b. At this point, the camera's view should be nearly centred left-to-right (within ~1 cm) on the virtual reality display. If not, then the screws that attach the camera assembly's hinge can be loosened in order to slightly shift the left-to-right angle of the camera head (one or both sides of the hinge can be used for this adjustment).
 - c. Re-check the vertical adjustment of the camera head.
4. Adjust the illuminator assembly angle:
 - a. Adjust the illuminator angle until the LEDs of the illuminator are visible via a double-reflection as per [Figure: 7.21](#).
The LEDs are quite dim, so it is recommended to close the vision blockers at this step and/or turn off the lights in the room.

Figure 7.21: View of Illuminator LEDs



- b. Tape a piece of plain white paper to the front of the virtual reality display so that it hangs down where the subject's head would normally be.
- c. Adjust the illuminator angle using the locking hinge until a horizontal shadow edge appears on the bottom of the EyeLink setup screen image, as per [Figure: 7.22](#).

Figure 7.22: Shadow Edge When Illuminator Angle Is Too High



- d. Lower the illuminator angle to the angle at which the edge of this shadow just begins to disappear. Do not over-lower the illuminator.
5. Adjust the camera focus:
 - a. Set up a subject as shown in the Kinarm Lab Operator Guide.
 - b. Using the small thumb screw on the camera lens, adjust the focus of the camera until the corneal reflection (bright dot) is as small and crisp as possible as shown in [Figure: 7.20](#). Tighten the thumb screw when the focus is set.

ATTENTION: Do not worry about the focus of the rest of the eye or the eye lashes - it is the corneal reflection that is important.

ATTENTION: Once adjusted, do not adjust the position, angle or focus of the camera and illuminator assemblies and avoid accidental bumping of these components to ensure that they remain in the correct position.

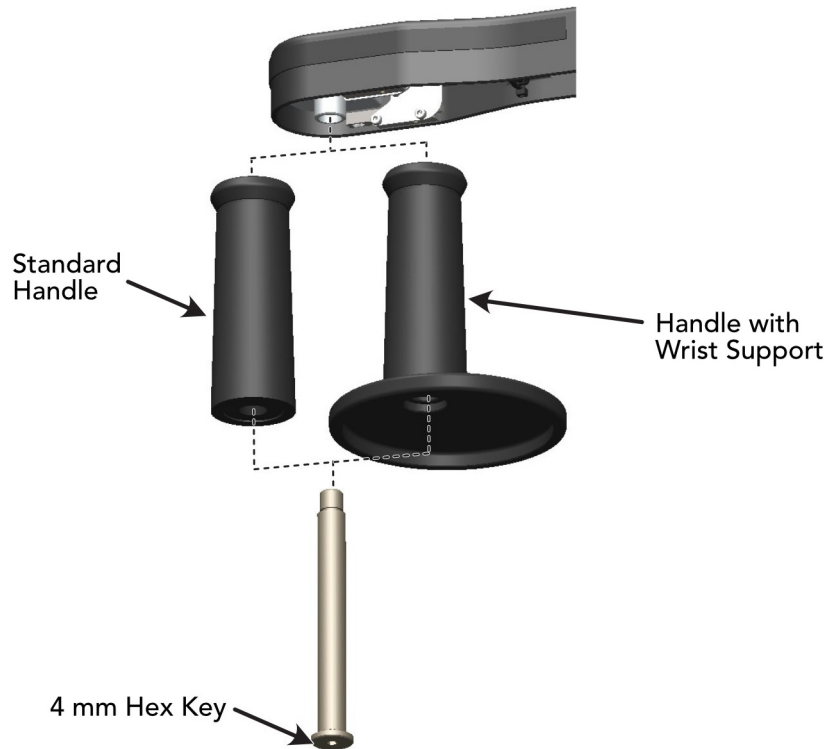
7.12 Install Handles

The following is required for installing a handle:

- 4 mm hex key
- Dexterit-E (for Handle Tightening when Force/Torque Sensors are installed)

The Kinarm End-Point Lab comes with a standard handle and/or an optional handle with wrist support. As shown in [Figure: 7.23](#), the two handle designs share the same mounting system. If it is necessary to install or swap these handles or perhaps a custom handle (see [10.3 Custom Handles](#)), use the following instructions to install the handles.

Figure 7.23: Install a Handle on the Robot



1. Loosen the handle shaft of the existing handle using the 4 mm hex key.
2. Remove the shaft from the handle core.
3. Insert the shaft into the other handle.
4. Install the new handle onto the robot.
 - If a Force/Torque Sensor is NOT installed, tighten the shaft.
 - If a Force/Torque Sensor is installed, use the handle tightening feature in Dexterit-E to assist with the installation to prevent undue torques on the sensor when tightening (see Dexterit-E User Guide for more details).

ATTENTION: The Force/Torque Sensor can be damaged if the handle is tightened with too much torque.

7.13 Verify the Kinarm Lab

When the Kinarm Lab is fully assembled and all wires and connections are complete, a verification process must be completed to ensure the Lab is operating as expected. Please

refer to the Kinarm Lab Operator Guide as needed for details on how to perform the steps below. Additional detail about the verification task itself can be found in the Dexterit-E User's Guide.

1. Power on the Lab.
2. In Dexterit-E, enter verification mode.
 - a. Navigate to **Kinarm Lab > Service Modes > Verification Mode**.
3. Calibrate the Lab (as applicable).
4. Select and run the **Verification** task for each robot.
5. Follow the Task Instructions.
6. Generate a verification report.
 - a. In the **Subjects** tab, select the data file you created and click the **Generate Report** button.
 - b. Confirm that the correct data file(s) are selected and click **OK**.
7. Review the report with Kinarm Support to ensure that the Kinarm Lab is working as expected.

8 Understanding the Kinarm Lab

This chapter of the guide explains some aspects or features of the Kinarm Lab in order to provide a better understanding of what to expect when using the Kinarm Lab.

8.1 Kinarm Lab Maintenance

The Kinarm Lab is largely maintenance free. Please refer to the Kinarm Lab Operator Guide for suggested maintenance checks.

8.2 Powering the Kinarm Lab Components

Power to all components of the Kinarm Lab is provided by the Power Console (mounted in the Power Rack). There are two types of power outlets on the Power Console: switched and always-powered.

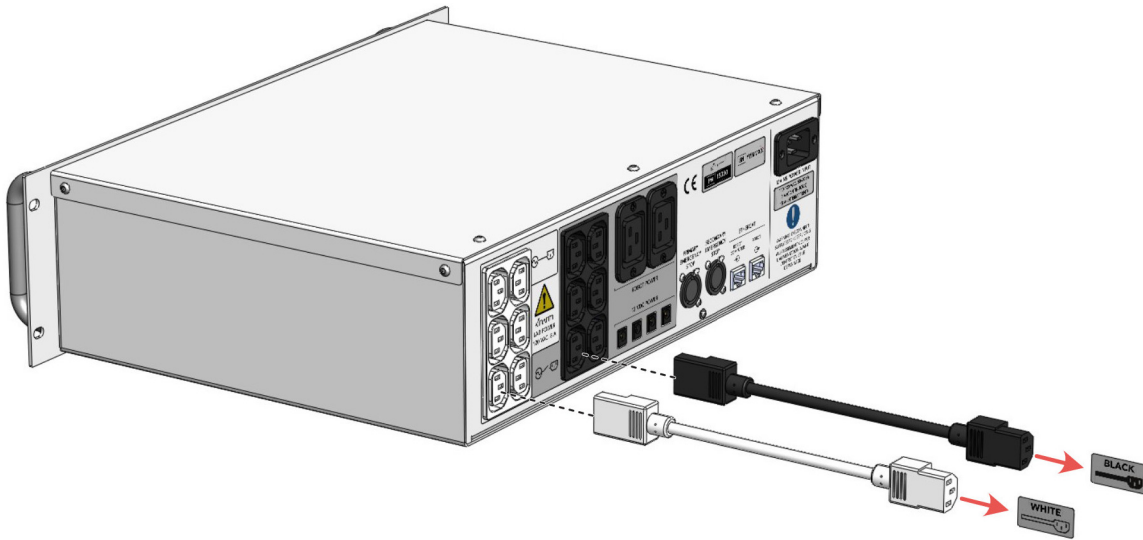
WARNING: To reduce the risk of electric shock, do not connect anything that is not part of the Kinarm Lab to the Power Console.

- **Switched.** All components plugged into the switched outlets are controlled by the on/off power switch on the front of the Power Console. In addition to regular outlet types, included in this selection of outlets are 2 special types: Robot Power Outlets (intended for Kinarm Robots only), and 12 VDC power (barrel connectors). These switched outlets are BLACK, and all power cables that are plugged into these outlets must also be BLACK. The following components must be plugged into the switched outlets:
 - Kinarm Robot(s) (Robot Power)
 - Speakers (12 VDC Power)
 - Robot Computer
 - AV Controller (12 VDC Power)
 - Subject Display
 - Gaze Power Box (if applicable - Kinarm Gaze Tracker only)
 - Gaze Base Unit (if applicable - Kinarm Gaze Tracker only)
- **Always-Powered.** All components plugged into the always-powered outlets will always have access to power (i.e. the Power Console's on/off power switch does not control power to these outlets). These always-powered outlets are WHITE, and all power cables that are plugged into these outlets must also be WHITE. The following components must be plugged into the always-powered outlets:
 - Dexter-E Computer
 - Operator Display
 - Network Switch

- Deluxe Chair charger
- Gaze-Tracker Computer (if applicable - Kinarm Gaze Tracker only)
- Gaze Display (if applicable - Kinarm Gaze Tracker only)

To assist with setup, each component that requires power has a label beside the power input port indicating which colour cable must be used (see [Figure: 8.1](#)).

Figure 8.1: Power Console Power Cable Colouring and Labeling

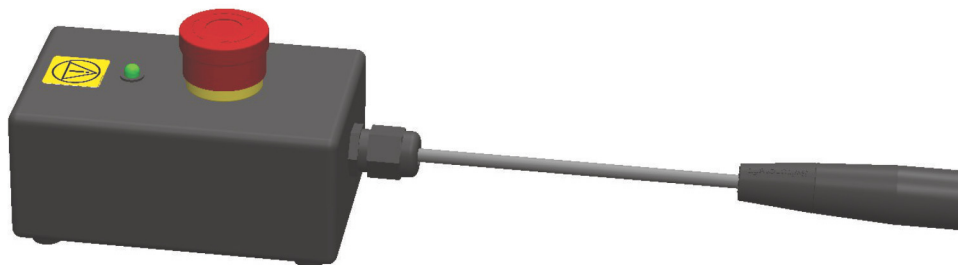


8.3 Emergency Stop

The Kinarm End-Point Lab comes with a hand-held emergency stop button. The emergency stop button must be connected and enabled for the motors to be enabled and produce forces.

The emergency stop button (see [Figure: 8.2](#)) connects to the primary emergency stop port on the Power Console, and serves to enable or disable all motors in the Kinarm End-Point Lab. The LED in the emergency stop button enclosure turns green when the button is enabled, and red when disabled. Push the button down to disable the motors. Pull the button up or rotate it until it pops back up to re-enable the motors.

Figure 8.2: Emergency Stop Button



It is possible to add a custom, secondary emergency stop button to the Kinarm Lab using the secondary emergency stop port located on the back of the Power Console. By default, the

secondary emergency stop port is bypassed and blocked by a cover. See [10.1 Enable Use of a Secondary Emergency Stop](#) to install a secondary emergency stop.

8.4 Grip Sensors and Active Braking

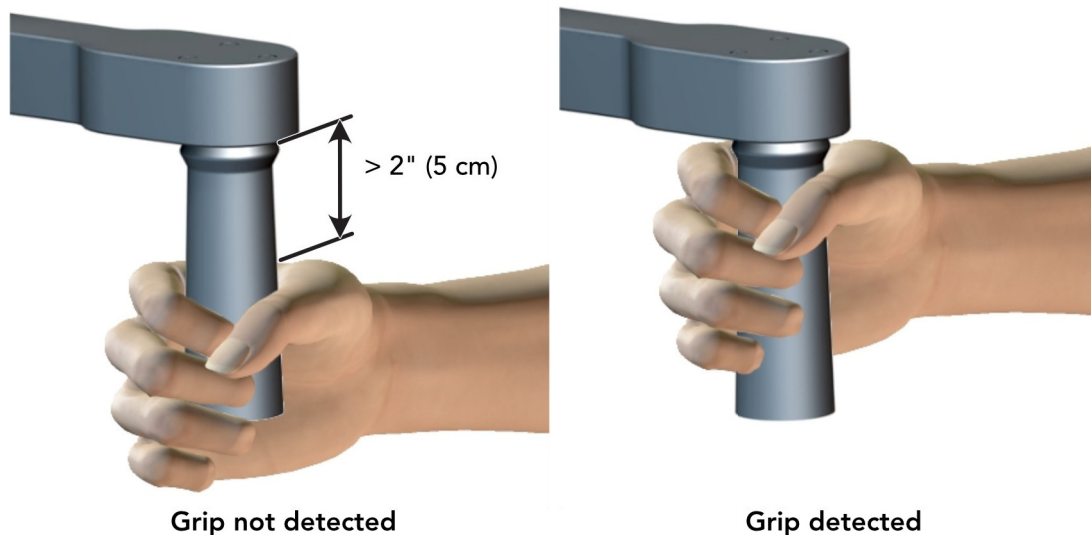
Sensors are built into each Kinarm End-Point Robot to detect if the handle is being properly grasped. If the grip sensors detect that the handle is not being grasped while a task is running, then active braking mode will be enabled. Active braking is similar to physical brakes in that the purpose is to stop movement. In this mode, a highly viscous force field is applied by the robot to quickly stop movement of the linkage. Active braking cannot be enabled unless the motors are active and able to produce forces (i.e. when Task Programs are running).

The grip sensors are embedded in the underside of Link L2, just above the handle. The sensors will be triggered when the handle is fully grasped, and when the subject's hand is within ~2" (5 cm) of the underside of Link L2 (see [Figure: 8.3](#)). The sensors in each robot act independently.

When the handle for a particular robot is grasped within the sensing range of the grip sensors, the status LED for the grip sensor on the back of the robot will switch to green as described in [9.1 Status Lights on the Robot](#). If the motors are active (solid green), then they will be switched to normal operating mode and will produce the forces commanded by the Task Program.

When the handle is released or not grasped properly, the status LED for the grip sensor on the back of the robot will switch to red as described in [9.1 Status Lights on the Robot](#). If the motors are active (solid green), then they will be switched to active braking mode.

Figure 8.3: Using the Handle Grip Sensors



8.5 Calibration of Kinarm Labs

There are two types of calibrations required by the Kinarm Lab: factory calibrations (performed by Kinarm prior to shipping) and subject calibrations (performed by the operator during normal use).

8.5.1 Factory Calibrations

The following one-time calibrations are completed by Kinarm during manufacture and/or installation of the Kinarm Lab. They may need to be re-done following servicing. Do not run any of the Factory Calibrations unless instructed by Kinarm Support.

Subject Display

This calibration ensures the subject display is correctly positioned and scaled relative to the components in the virtual reality display. Refer to the Dexterit-E User Guide for details on how to perform this calibration.

Robot Axis Position

This calibration locates the robots relative to the lab workspace. Refer to the Dexterit-E User Guide for details on how to perform this calibration.

Absolute Encoder Offset

The absolute encoder offset calibration precisely identifies the relative angle of the robot joints based on the known position of the linkage when positioned with the calibration bar. Please refer to the Dexterit-E User Guide for details on how to perform this calibration.

8.5.2 Subject Calibrations

The following calibrations must be completed for each subject.

Force/Torque Sensor Zeroing

If force/torque sensors are installed, they will need to be zeroed with each subject to account for drift. Refer to the Kinarm Lab Operator Guide for details on performing the calibration.

Kinarm Gaze-Tracker Calibration

The Gaze-Tracker calibration procedure maps the subject gaze vector to the Virtual Reality Display workspace. This calibration must be performed for each subject and each session. If the

subject's target sticker is moved then the Gaze-Tracker will have to be re-calibrated. Refer to the Kinarm Lab Operator's Guide for details on how to perform this calibration.

8.6 Adjustment and Operation of the Deluxe Chair

To comfortably fit a wide range of subjects, the deluxe chair is adjustable in a variety of ways. This section explains, in general, how to make these adjustments.

8.6.1 Adjust the Seat Height of the Deluxe Chair

The seat height of the Deluxe Chair is adjusted using the wired remote control. [Figure: 8.4](#) shows which buttons can be used for up and down movement. The row of LED indicators at the top of the unit have the following meanings:

- **Left LED (red):** If lit, this LED indicates that the battery is in need of charging. The chair will not operate in this state in order to protect the health of the battery.
- **Centre LEDs (green):** These three LEDs together indicate the level of charge of the battery. All three lit means full charge. As the charge depletes, the LEDs turn off from right to left. When the charger is plugged in, these LEDs will incrementally flash from left to right, indicating that the charge level is increasing.
- **Right LED (red):** If lit, there is an error with the chair battery or controller.

ATTENTION: The height cannot be adjusted manually or if the battery is not charged.

ATTENTION: The height adjustment does not work while the charger is plugged into the chair.

The wired remote control can be stored on the storage bracket on the rear of the backrest, and is held in place by a magnet that is built into the top of the wired remote control. This magnet also allows the wired remote control to be stored anywhere on the chair frame when not in use.

Figure 8.4: Wired Remote Control

8.6.2 Armrest Adjustments for the Deluxe Chair

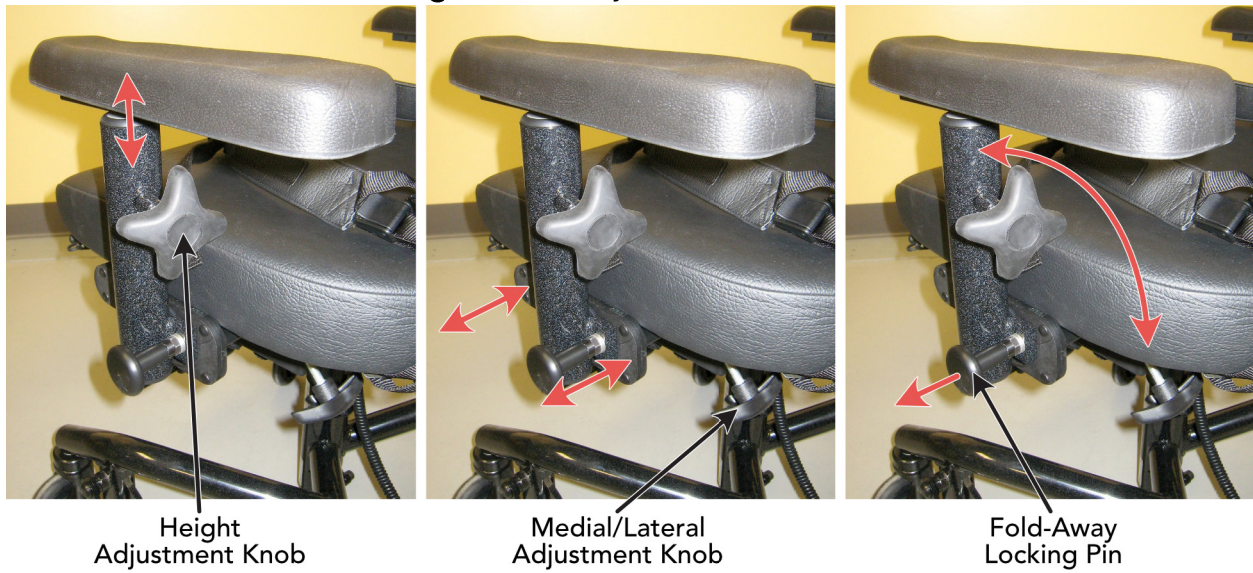
The Deluxe Chair armrests are adjustable in several ways. The following describes how to make the adjustment, and [Figure: 8.5](#) highlights the locations of the adjustment knobs.

- **Height.** Each armrest has approximately 4" (10 cm) range of height adjustment. To change the height, loosen the knob, move the armrest to the desired height, and tighten the knob.

NOTE: The armrest height adjustment mechanism is tapered, which means that the knob must be loosened more as the armrest is moved further down. If moving the armrest up from the lowest position to the highest position, the knob will need to be turned several times before it will begin clamping.

- **Medial/Lateral.** Each armrest has approximately 1" (2.5 cm) range of medial/lateral adjustment. To change the position, loosen the knob, move the entire armrest to the desired position and tighten the knob.
- **Fold Away.** Each armrest can be folded away in order to provide easy access for subject transfer and to avoid interference with the Kinarm End-Point Lab during exams. To fold the armrest away, pull the fold-away locking pin, and pivot the armrest backwards. To bring the armrest back up, lift the armrest, and ensure that the fold-away locking pin locks back into place.

Figure 8.5: Adjust the Armrests



8.6.3 Footrest Adjustments for the Deluxe Chair

The Deluxe Chair footrest is adjustable. The following describes how to make the adjustment, and [Figure: 8.6](#) highlights the location of the adjustment knob.

- Height.** The footrest can be adjusted up and down over a range of approximately 6" (15 cm) in 0.75" (20 mm) increments to accommodate subjects of different leg length. To change the height, loosen the knob slightly, move the footrest to the next position, and once in position, tighten the knob. The positions are most easily felt by loosening the knob only slightly. As the footrest is moved up or down, the increments can be felt as notches in the footrest rub against the knob's screw.

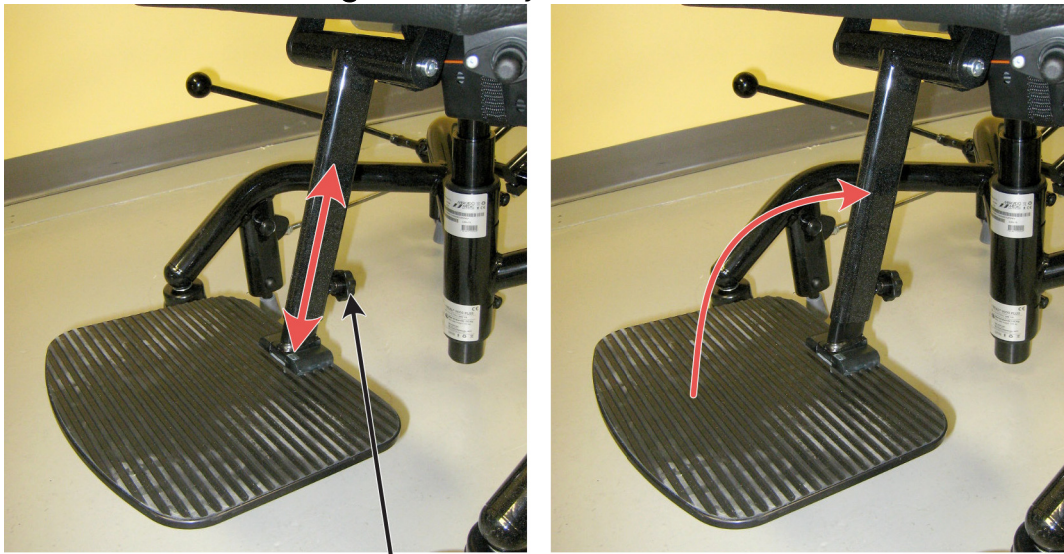
CAUTION: Do not extend the footrest beyond the minimum insertion of ~1.75" (4.5 cm), as indicated by the label on the footrest post. The label is exposed only if the footrest does not meet the minimum insertion.

CAUTION: Be sure to tighten the knob after an adjustment. If the footrest is loose, it could fall out when under load, causing a subject to fall.

ATTENTION: If the chair is in the lowest position, the footrest will touch the floor before reaching the longest setting. In this case, the chair will need to be raised before the full adjustment can be made. Likewise, if the footrest is adjusted to the longest position, it could hit the floor when the chair is lowered.

- Fold Away.** The footrest can be folded up under the chair when not in use. There are no knobs or clamps - simply lift or pull down the footrest platform as needed.

CAUTION: When the footrest is folded up, ensure that the subject's heels are not underneath the footrest prior to lowering the chair, otherwise the subject's heels could be caught underneath.

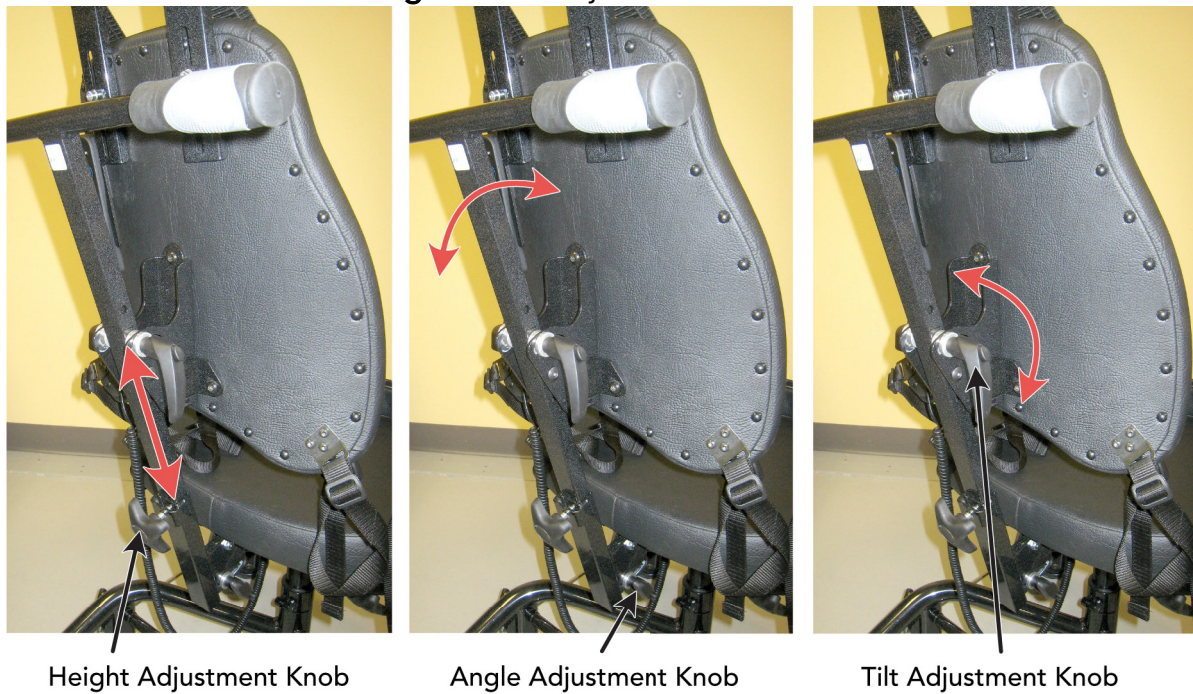
Figure 8.6: Adjust the Footrest

Height Adjustment Knob

8.6.4 Backrest Adjustments for the Deluxe Chair

The Deluxe Chair backrest is adjustable in several ways. The following describes how to make the adjustments, and [Figure: 8.7](#) highlights the locations of the adjustment knobs.

- **Height.** The backrest has [Figure: 8.7](#) approximately 5" (12.5 cm) range of height adjustment. To change the height, loosen the knob, move the backrest to the desired height, and tighten the knob.
- **Angle.** The backrest mechanism angle can be adjusted over approximately 45 degrees (20 degrees forward and 25 degrees backwards) to bring the entire backrest forward or backward relative to the seat. To change the position, loosen the knob, move the entire backrest mechanism to the desired position and tighten the knob.
- **Tilt.** The backrest itself can be tilted by approximately 45 degrees so that the backrest can be comfortably oriented for any given backrest mechanism angle (see above). To change the position, loosen the knob, pivot the backrest, and tighten the knob.

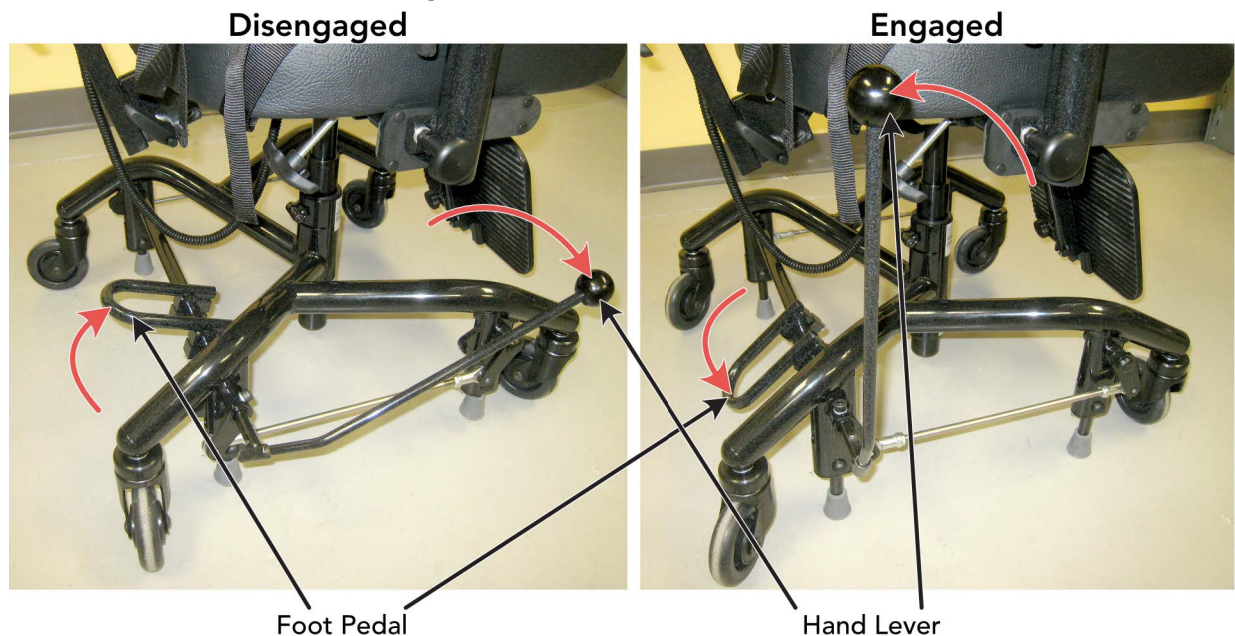
Figure 8.7: Adjust the Backrest

8.6.5 Deluxe Chair Brakes

The brakes of the Deluxe Chair consist of four spring-loaded feet that are pushed against the floor when the brakes are engaged. The foot pedal is for use by the operator, and the hand lever can be used by either the subject or the operator. The hand lever can be placed on either side of the chair. Refer to [Figure: 8.8](#) for details on brake operation.

- **Disengaged.** The brakes are disengaged by either pressing the hand lever forward into a nearly horizontal position OR by raising up the foot pedal.
- **Engaged.** The brakes are engaged by either pulling the hand lever up into a nearly vertical position OR by pressing down on the foot pedal.

Figure 8.8: Operate the Brakes

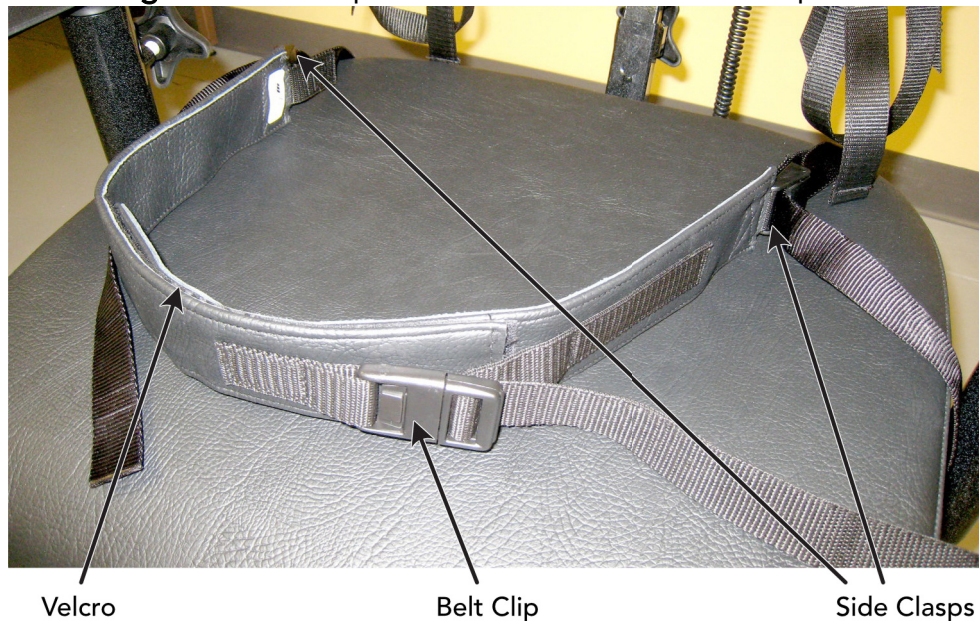


8.6.6 Adjust the Deluxe Chair Hip Belt

The subject hip belt is a single strap belt that splits into two halves at the centre. It helps to maintain the position of the subject during an exam. The following identifies the main components and how to properly use them. See [Figure: 8.9](#).

- **Velcro and Belt Clip.** The Velcro and belt clip connect the two halves of the hip-belt. There is approximately 8" (20 cm) of adjustment range in the central connection of the belt. The Velcro keeps the belt in place until the belt clip is properly attached.
- **Side Clasps.** The side clasps provide an additional 30" (76 cm) of adjustment range for the belt if necessary. The clasps can be opened, the strap pulled through to change the length, and then closed again.

Figure 8.9: Components of the Deluxe Chair Hip Belt



8.6.7 Adjust the Deluxe Chair Harness

The subject harness is an H-Strap style harness that helps to maintain the position of the subject during an exam. The subject harness brackets can be used to provide extra range of adjustment, if needed. The following identifies the main components and how to work them correctly. See [Figure: 8.10](#).

- **Chest Clip.** The chest clip connects across the front of the subject's chest and can be adjusted in length.
- **Shoulder Clasps.** The shoulder clasps allow the harness to be adjusted higher or lower on the subject's shoulders, as needed, for comfort. The clasps can be opened, the strap pulled through to change the length, and then closed again.
- **Side Clips.** The side clips are used to tighten the harness against the subject. If necessary, they can be unclipped to set up or remove a subject from the chair.
- **Harness Brackets.** The subject harness brackets can be adjusted in height over a range of approximately 3" (7.5 cm) to accommodate different subject heights. The brackets can also be angled laterally by up to approximately 30° to provide a wider connection over the shoulders.
 - To adjust the height, use a 5 mm hex key (provided with the chair) to loosen the top two screws, slide the bracket up or down to the desired location, and tighten both screws.
 - To adjust the angle, raise the bracket to its highest location, tighten the lower of the two screws, loosen the third screw (lowest) that is exposed when the bracket is

raised, twist the bracket to the desired angle, and tighten the screws. Set the height of the bracket to the desired height.

NOTE: The angle adjustment should rarely be needed, if at all. The screws may need to be loosened several turns before the bracket will rotate smoothly. The metal bracket is tacked to the backrest material and the curved contours of the backrest may prevent smooth rotation unless the screws are quite loose.

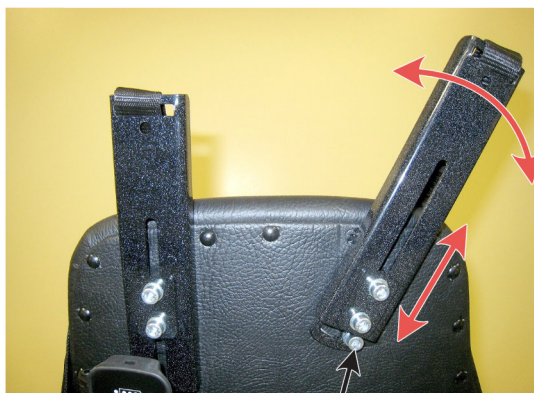
Figure 8.10: Components of the Deluxe Chair Harness



Standard Harness Bracket Setup



Angled Harness Bracket Setup



8.7 Recording Timing of Visual Stimuli (120 Hz Subject Display)

The optional 120 Hz subject display has a feature called Pixel Mode which configures the monitor to output the RGB value of the top-left pixel onto the digital output of the monitor. This feature is enabled by default on all Kinarm Labs with this display.

NOTE: This feature is not available on the standard 60 Hz subject displays.

Pixel mode configures the display to output the RGB value of the top-left pixel onto the digital output on the back of the display (see [9.6 120 Hz Subject Display Digital Outputs](#)). There is 1 μ s delay between the timing change of that digital output and a change to that pixel. What the subject sees will be somewhat delayed by the response time of that pixel (nominally 10 ms grey-to-grey), but it begins to change nearly instantaneously. As with all scanning displays it takes a full frame (i.e. 8.33 ms @ 120 Hz) to scan from that first top-left pixel to the last bottom-right pixel. The timing of visual stimuli elsewhere on the screen is therefore location dependent, but entirely deterministic (i.e. for the i^{th} row, it will be $(i/1080) * 8.33$ ms after the digital output).

To take advantage of this feature, the end-user will have to manufacture a custom cable (not provided by Kinarm) to connect the digital output of the display to the digital inputs of the Kinarm Lab's DAQ. Additionally, custom task programs will need to be updated to acquire these digital signals.

8.8 Speaker Volume Adjustment

There are two different control approaches to adjust the volume of the audio output of the speakers in the Virtual Reality Display, which serve two different purposes.

- One-time setup control (i.e. during installation)
 - The amplifier driving the speakers has a built in volume knob. Turn the knob clock-wise to increase the volume, and counter-clockwise to decrease the volume.

NOTE: The amplifier volume knob is normally not adjusted. It is set to the maximum at the factory and re-confirmed during installation.

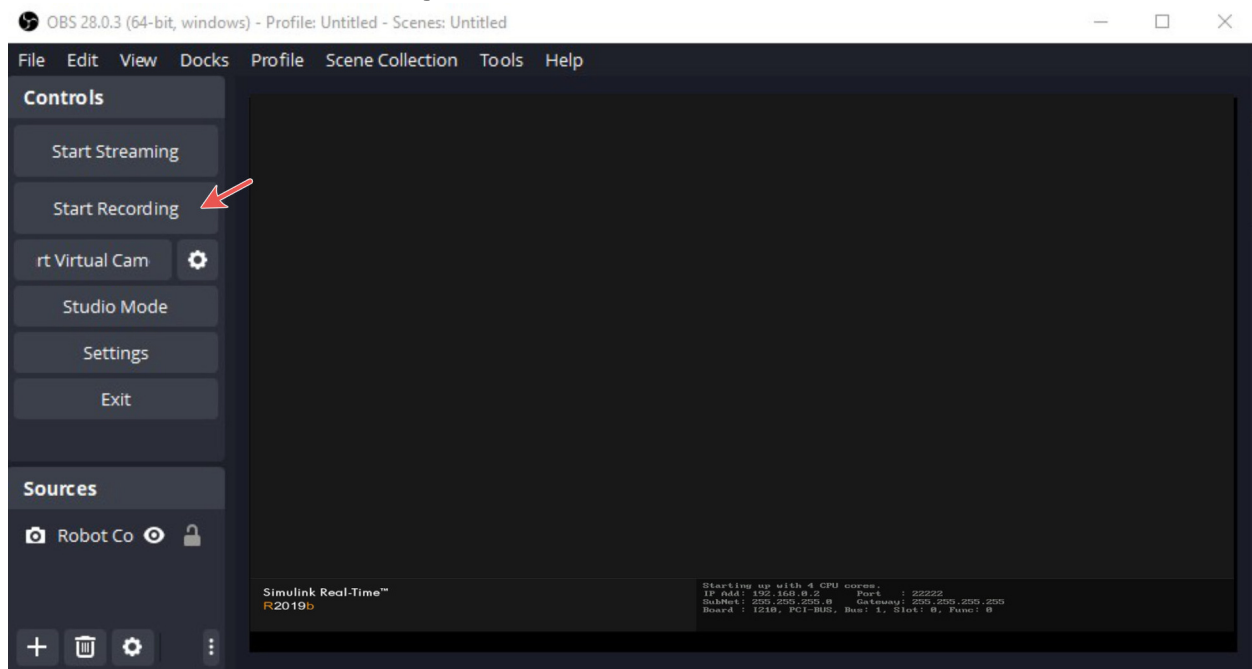
- Ongoing adjustments (i.e. during normal operation)
 - There are a couple of different means to adjust speaker volume directly through Dexter-E. Please refer to the Kinarm Lab Operator Guide or the Dexter-E User Guide for more details.

8.9 Robot Computer Video Output

While the video display of the Robot Computer is often not needed when collecting data, it can be very important during custom task development and troubleshooting. Kinarm Labs come with a screen capture device and software for displaying and recording this video.

The screen capture hardware streams the video output of the Robot Computer to the Dexterit-E Computer via USB. OBS Studio software installed on the Dexterit-E Computer (pinned to the Windows taskbar) displays the video (see [Figure: 8.9](#)).

Figure 8.11: OBS Studio Window



The following are some useful tips for using this software.

- The OBS Studio software window can be resized to any size, scaling the Robot Computer video to fit the window.
- For the most crisp view of the video, maximize the OBS Studio window to full screen
- To record the video, press the **Start Recording** button on the left hand side of the screen. The button will change to **Stop Recording**, which can then be clicked when needed to stop the recording.
- The recorded video will be automatically saved to the desktop on the Dexterit-E Computer.

NOTE: When developing custom tasks, it is often desirable to add a separate display for the Robot Computer. The screen capture hardware has an HDMI output port that can be used to connect a separate display while maintaining the functionality described above.

8.10 Moving the Kinarm Lab

Contact Kinarm Support if it becomes necessary to move the Kinarm Lab. Many of the disassembly steps will be the reverse of the steps outlined in [7 Unpack and Assemble](#). However, the steps may be different depending on whether the Kinarm Lab is being moved to a different location in the same room, to a new room in the same building, to a new building,

or even a new city. Kinarm Support will assist you with determining the safest way to move the Kinarm Lab to the new location.

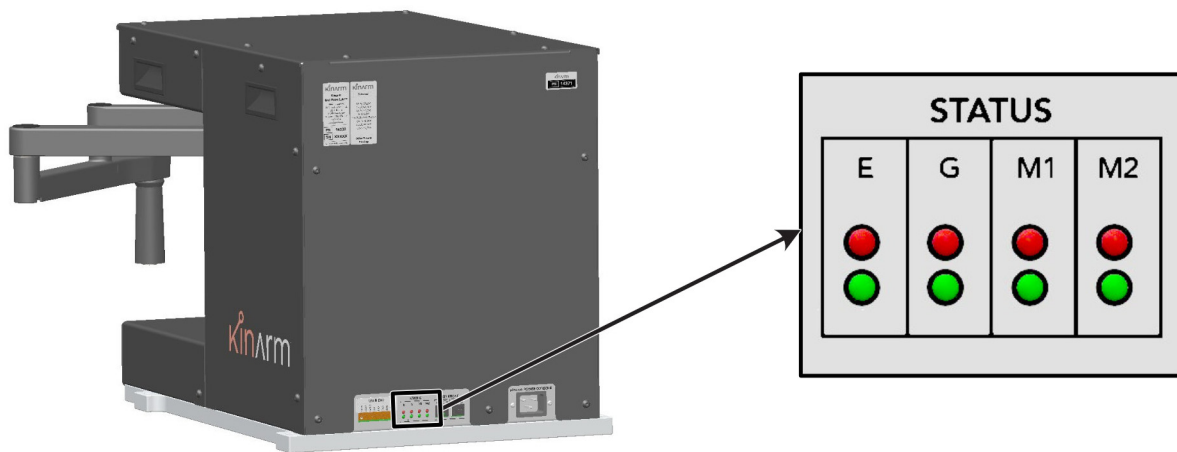
9 Kinarm Lab Reference

The sections in this chapter provide details about the Kinarm Lab that may be a useful reference for using the Lab. This includes specification of status indicators, dimensions, pinouts and settings that may be necessary to know when working with the Lab.

9.1 Status Lights on the Robot

On the back of each Kinarm End-Point robot is a grid of eight of status LEDs with red on the top row and green on the bottom row. When first powered on, all eight LEDs simultaneously light up. After 2-3 seconds the LEDs begin to indicate the present status of the robot. The meaning of each is summarized below and in [Figure: 9.1](#).

Figure 9.1: Status Lights on the Robot



- Emergency Stop Status LEDs: E
 - **RED (Solid):** Indicates that the stop is depressed or disconnected and therefore the robot will not produce forces.
 - **GREEN (Solid):** Indicates that the stop is not depressed and therefore the robot can produce forces, if the motor status LEDs are solid green.
- Grip Sensor Status LEDs: G
 - **RED (Solid):** Indicates that grip is not sensed. If the motors are active (solid green), then the they will be switched to active braking mode (see [8.4 Grip Sensors and Active Braking](#)).
 - **GREEN (Solid):** Indicates that the grip has been sensed. If the motors are active (solid green), then they will be switched to normal operating mode and will produce the forces commanded by the Task Program.

- Motor Status LEDs: M1 and M2
 - **RED (Solid):** Indicates a fault was found on the motor or that the emergency stop button is pressed. All motors will be red if the emergency stop is pressed. Data collection can proceed with a faulted motor, but forces will not be produced.
 - **GREEN (Solid):** Indicates that the motor is active and able to produce forces.
 - **GREEN (Flashing):** Indicates that the motor is waiting and ready to work if a task starts, but currently will not produce forces. The robots can produce forces only when a task has been started in Dexter-E and this LED turns to solid green.

9.2 Robot User Digital Inputs and Outputs (DIO)

In addition to the optional DAQ modules, the Kinarm End-Point Lab has several electrically isolated digital inputs and outputs available for users on the Kinarm robots (accessible through the Dexter-E TDK). The connector, which can be located on the back of each robot, is an 8-pin connector with pinouts as shown in [Figure 9.2](#). An external mating connector is required (Weidmuller 1627300000 or equivalent). The pinout functions are shown in [Table 9.1](#).

NOTE: The +5V and GND pins as well as all inputs and outputs are from an electrically isolated supply. All inputs and outputs are TTL compatible.

Figure 9.2: User DIO Pinout Diagram

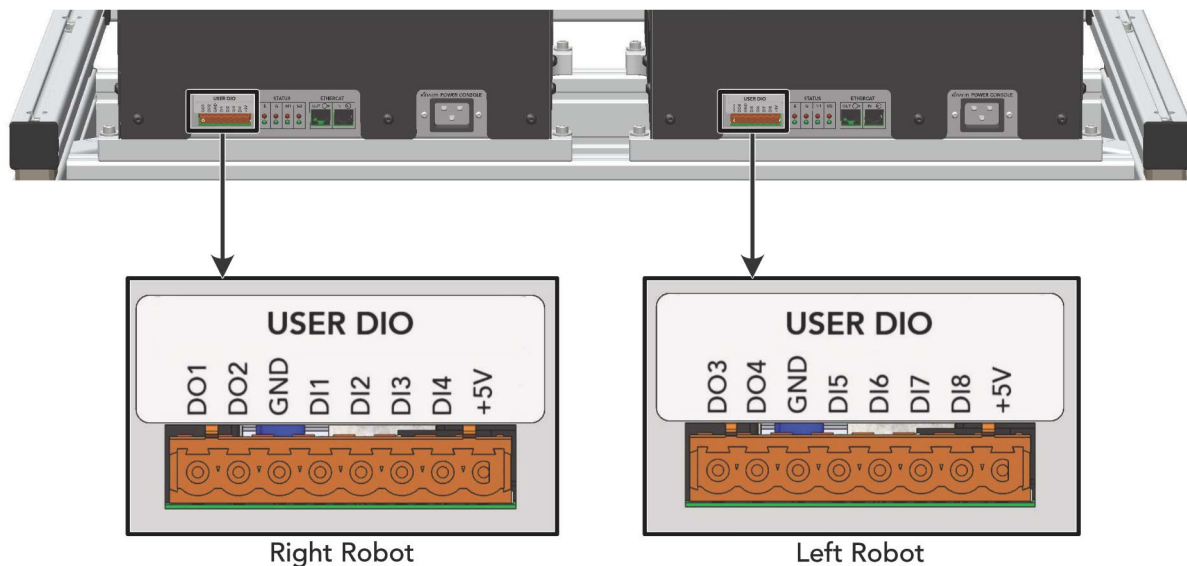


Table 9.1: User DIO Pinout Functions (Sheet 1 of 2)

Pin Number	Right Robot Function	Left Robot Function
1	DO1	DO3
2	DO2	DO4

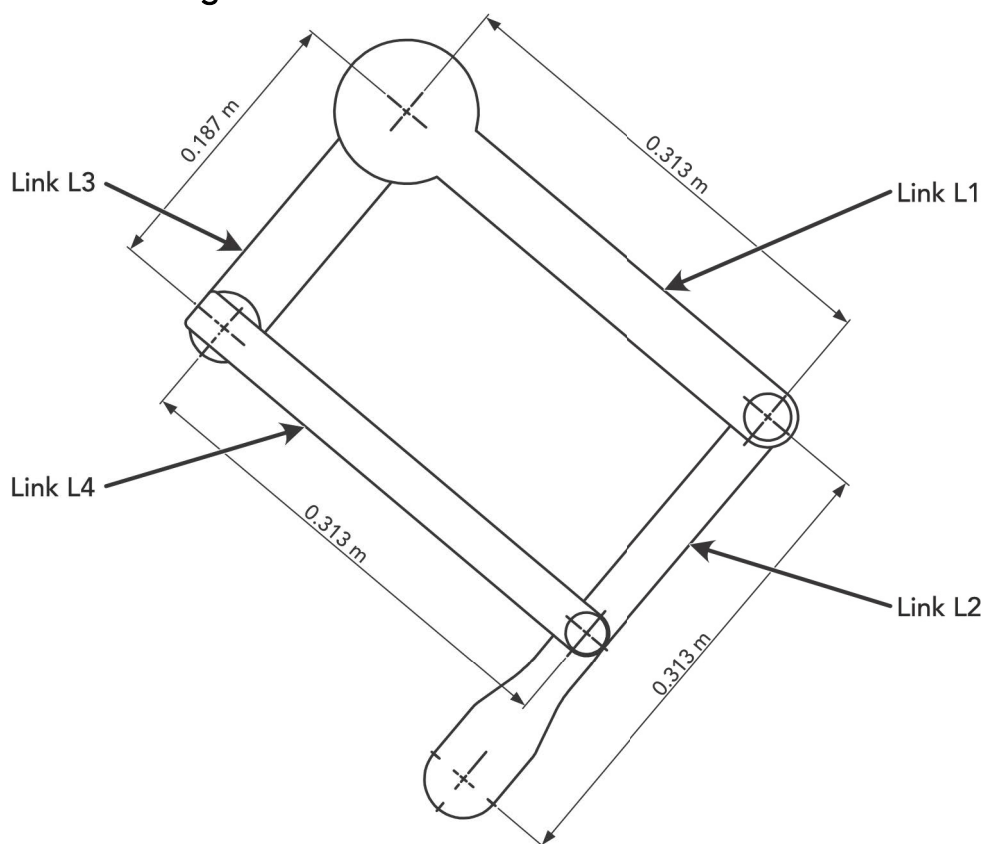
Table 9.1: User DIO Pinout Functions (Sheet 2 of 2)

Pin Number	Right Robot Function	Left Robot Function
3	GND	GND
4	DI1	DI5
5	DI2	DI6
6	DI3	DI7
7	DI4	DI8
8	+5V	+5V

9.3 Robot Link Definitions and Dimensions

The Kinarm End-Point robot links are defined in [Figure: 9.3](#). The lengths of the links are constant and cannot be changed.

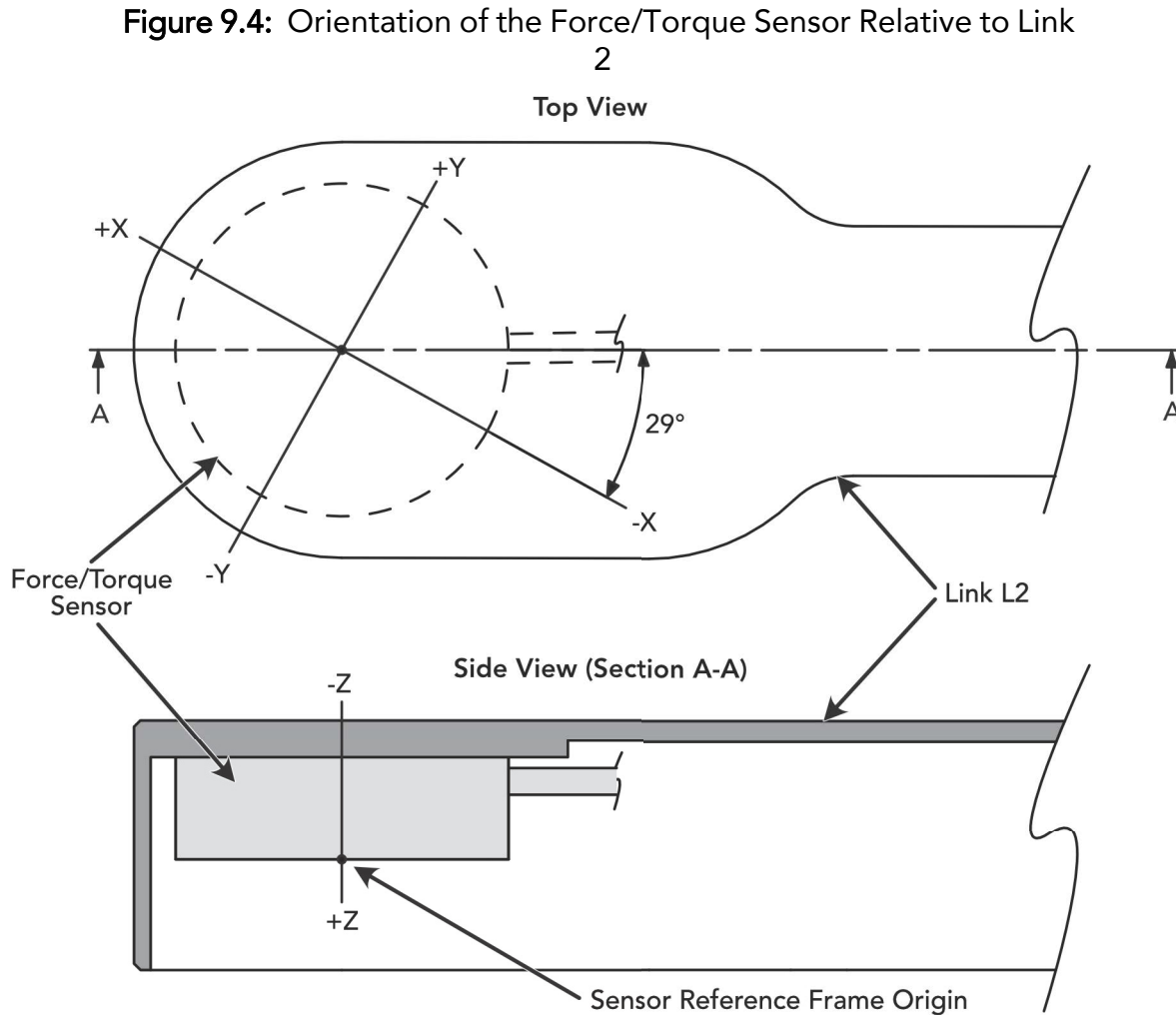
Figure 9.3: Link Definitions and Dimensions



9.4 Force/Torque Sensor Frame Orientation

The force/torque sensors are mounted into Link 2 of the Kinarm End-Point robot as shown in [Figure: 9.4](#). The orientation of the sensor relative to Link 2 is the same for both right and left configurations of the robot.

Kinarm's Dexter-E software automatically maps the sensor output to the coordinates of the lab workspace for the Kinarm End-Point Labs. However, if Dexter-E is not used, the force/torque sensor output must be manually mapped to the desired coordinates. [Figure: 9.4](#) can be used to make this mapping.



9.5 Force/Torque Sensor Filter Settings

The force/torque sensor settings include an internal filter for the force/torque data stream. This filter is a first order low pass filter with 326 Hz cutoff frequency. The delay added by this filter is ~0.5 ms. Contact Kinarm Support if it is important to change or remove this filter.

9.6 120 Hz Subject Display Digital Outputs

The 120 Hz Subject Display has digital outputs that can be connected to the digital inputs of the Kinarm Lab's DAQ in order to record the timing of visual stimuli (see [8.7 Recording Timing of Visual Stimuli \(120 Hz Subject Display\)](#) for explanation of the feature). [Figure: 9.5](#) and [Table: 9.2](#) below specify the pinouts and mapping necessary to use this feature.

Figure 9.5: Digital Output Pinout Diagram of Connector on Back of Display

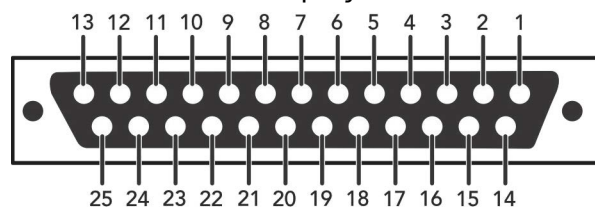


Table 9.2: Mapping from D-Sub pin to RGB bit

Pin (Top row of D-Sub)	RGB bit	Pin (Bottom row of D-sub)	RGB bit
1	B0	14	B1
2	B2	15	B3
3	B4	16	B5
4	B6	17	B7
5	G0	18	G1
6	G2	19	G3
7	G4	20	G5
8	G6	21	G7
9	R0	22	R1
10	R2	23	R3
11	R4	24	R5
12	R6	25	R7
13	N/A	Shield	N/A

10 Customizing the Kinarm Lab

The Kinarm Lab offers some ways to customize the hardware. The following sections provide some guidance for some common customizations for the Lab. Contact Kinarm Support for guidance.

10.1 Enable Use of a Secondary Emergency Stop

It is possible to add a secondary emergency stop button to the Kinarm Lab using the secondary emergency stop port located on the back of the Power Console. When both primary and secondary emergency stops are in use, all motors are disabled if either one of the buttons are pressed. To re-enable the motors, both buttons must be released. The LED in the emergency stop button enclosures are independent from each other, meaning that the colour for each button does not reflect the status of the other button.

ATTENTION: Dexterit-E does not distinguish between the two emergency stops. This means that the emergency stop status indicator in Dexterit-E will be red when either stop is pressed. Additionally, the data file records only that an emergency stop has been pressed, and cannot identify which button is pressed.

The following is required for configuring the Kinarm Lab to work with a secondary emergency stop:

- 2.5 mm and 3 mm hex keys
- Needle-nose pliers (optional)

Instructions:

1. Disconnect all wires from the back of the Power Console.
2. Remove the Power Console from the Power Rack:
 - a. Remove the 4 screws (with plastic washers) that attach the Power Console to the Power rack
 - b. Pull the Power Console out from the Power Rack and place on the floor or a work table.

ATTENTION: The Power Console is heavy. Take care when lifting it out of the rack.
3. Remove the cover of the Power Console:
 - a. Remove the 7 screws that attach the cover to the enclosure
4. Reposition the jumper on the PCB
 - a. Locate the **ESTOP_P2 BYPASS** jumper on the PCB
By default, the jumper will be positioned on pins 1 and 2.

- b. Remove the jumper and reinstall it on pins 2 and 3.
5. Reinstall the Power Console cover.
6. Reinstall the Power Console into the Power Rack.
7. Remove the cap blocking the secondary emergency stop port.
The cap is attached to the enclosure by a lanyard, so it will dangle.
8. Plug the secondary emergency stop button into the secondary emergency stop port on the Power Console.

10.2 Custom Emergency Stop

The hand-held emergency stop button can be replaced with third-party switches if so desired. Likewise, third-party switches can be used as the secondary emergency stop. The required connector is a Switchcraft AA3M (Nickel Plated) or AA3MB (Black). The pin diagram is shown in [Table 10.1](#).

Table 10.1: Emergency Stop Connector Pin Diagram

Pin Number	Signal	Notes
1	GND	
2	+24 V	
3	RETURN	• GND = Disable • +24 V = Enable
Switchcraft AA3M (Nickel Plated) or AA3MB (Black).		

10.3 Custom Handles

The Kinarm End-Point Robot has a removable handle that can be replaced with a custom handle. The following are some tips and consideration for creating a handle design. Contact Kinarm Support to confirm that the custom handle design is appropriate for use with the Kinarm End-Point Robot.

For example, Arsalis has developed a custom handle called KINGRIP that provides a way to measure grip force. Details can be found at:

- <https://www.arsalis.com/products/instrumented-handles/>

10.3.1 Mounting Considerations

- The handle mounting plate in the robot linkage accepts a shaft with M10x1.5 thread. The maximum depth is 12.5 mm.

10.3.2 Controllability

The mass and inertia of the handle can affect controllability of the robot. Any force-fields that depend on feedback (e.g. position controller or active braking) may be adversely affected.

10.3.3 Force/Torque Sensor Considerations

If force/torque sensors are installed in the robot, it is imperative to take care to avoid applying undue forces and torques on the sensor, otherwise the sensor will be damaged.

- The single-axis overload limit for the ATI Mini-40 in the XY Torque direction is 19 N·m (see [9.4 Force/Torque Sensor Frame Orientation](#) for details about sensor reference frame). Any length added to the standard handle will increase the chance of overloading the sensor.
- The handle shafts included as part of factory handles are specifically designed to break before the force/torque sensor overload limit under certain load conditions. This should be a consideration when designing custom handles.
- Use the handle tightening feature in Dexterit-E to assist with the installation of the handle to prevent undue torques on the handle when tightening (see Dexterit-E User Guide for more details).

10.3.4 Grip Sensors Considerations

The grip sensors are not part of the handle itself (they are mounted in the robot linkage), but they are designed to work with the handle provided by Kinarm. Changing the dimensions of the handle may reduce the effectiveness of the grip sensors. Furthermore, depending on the handle design, the grip sensors may either be prevented from detecting grip at all, or cause them to falsely detect grip even when there is no hand grasping the handle.

CAUTION: It is important to note if a custom handle causes the grip sensors to falsely detect grip, the motors will not be able to switch to active braking mode if the handle is released. In this case, all forces commanded by the Task Program will continue to be applied unless the emergency stop is pressed.

11 Troubleshooting

11.1 All Motor Status LEDs Are Red

1. Check the status of the emergency stop status LEDs on the back of the robot(s). Do not look at the LED on the desktop emergency stop button itself, as it is the status LEDs on the back of the robot(s) that matter.
 - a. If the robot emergency stop status LEDs are red on both robots, then that is what is preventing the motors from enabling. See [9.1 Status Lights on the Robot](#) to determine which LED refers to the emergency stop. Proceed to the section [11.3 Emergency Stop Status LEDs on Robot Electronics Panel are Red](#).
 - b. If the emergency status LEDs are green, then proceed to the next step.
2. Check to see if Dexterit-E is started and connected to the Robot Computer.

NOTE: Dexterit-E provides a warning only when it cannot connect to the Robot Computer. If no warning is provided, Dexterit-E has successfully connected to the Robot Computer.

- a. If Dexterit-E is started and connected to the Robot Computer, and both emergency stop status LEDs are green, then try turning everything off and then on again to see if the problem persists. If it does, then contact Kinarm Support.

11.2 One or More (But Not All) Motor Status Lights Are Red

If one or more of the motors will not enable (but at least one other motor is enabled), turn the Power Console and the Robot Computer off. Power everything back on, and if the problem persists, contact Kinarm Support.

11.3 Emergency Stop Status LEDs on Robot Electronics Panel are Red

1. Check that the hand-held emergency stop button is plugged into the Primary Emergency Stop port of the Power Console.

This input needs to have a button plugged in for the emergency stop status LED on the robot electronics panel to be green.
2. Check if either of the emergency stops are depressed.

Twist the button or pull it up to 'pop' it up for enabling.
3. If neither of these suggestions fix the problem, contact Kinarm Support.

11.4 Emergency Stop Status LED on Hand-Held Emergency Stop is Red

1. Check if the emergency stop button is depressed.
 - If the button is down, twist the button or pull it up to 'pop' it up for enabling.
 - If the button is popped up and the status on the button remains red, contact Kinarm Support.

11.5 Kinarm Gaze-Tracker Power Box LED is Red

When the Gaze-Tracker Power box LED is red, it means that the gaze power box is powered, but the illuminator is not being powered because the fan is not providing sufficient cooling.

1. Without bumping it, check to feel if there is air flow around the fan.
2. If there is no air flow, unplug the fan power cable, and then plug it back in. Does the fan or LED change?
3. If the above suggestions do not fix the problem, contact Kinarm Support.

11.6 Kinarm Gaze-Tracker Camera Not Detected

When starting up the Gaze-Tracker Computer, and the Kinarm Gaze-Tracker camera is not detected, the Gaze-Tracker Computer will not complete its startup.

If this or another error is reported, there are a few steps that might resolve the problem.

1. Ensure the camera is powered on.

If the camera does not have power, the Gaze-Tracker Computer will display the following error: `ERROR:-89: No Optilink head found, check link or power`. Proceed with the following steps:

 - a. If the Gaze Power Box is not on, power it on.
 - b. If the Gaze Power Box is already on, cycle the power.
 - c. Reboot the Gaze-Tracker Computer.
2. Ensure the EyeLink base unit is enabled.

The base unit has an LED that indicates if it is enabled. If the base unit has entered into a disabled state, its LED may be amber instead of green. Cycling power to the base unit may reset this.

 - a. Unplug the power from the base unit.
 - b. Disconnect and reconnect the network cable from the base unit.

- c. Plug the power back into the base unit.
The LED on the base unit should turn green.
- d. Reboot the Gaze-Tracker Computer.

11.7 Deluxe Chair Seat Does Not Move Up or Down

There are several situations that might arise in which the seat will not move up or down.

Table 11.1: Deluxe Chair Seat Troubleshooting (Sheet 1 of 2)

Possible Causes	Recommended Solution
Battery not charged	Charge the battery. If the left LED on the wired remote control is lit (red) and/or an audible signal is emitted from the motor control box (under the chair), this indicates that the battery requires charging. ATTENTION: Completely discharging the battery will shorten the life of the battery.
Battery in stand-by mode	Charge the battery for at least 5 seconds, and then unplug the charger. If the left LED on the wired remote is not lit, and the centre LEDs on the wired remote flash green, this indicates that the battery has entered stand-by mode. NOTE: The battery enters standby mode after 7 days of non-use if not plugged in. Charging the battery for a few seconds and then unplugging it will re-activate the battery.
Battery dead	If the battery has been completely discharged (left unplugged, in stand-by mode for a long time without charging), the battery will not be able to be charged. The battery will need to be replaced. Contact Kinarm Support for assistance. To avoid this situation, the battery should be charged regularly. NOTE: The motor in the chair may make some clicking noises when attempting to move the chair up or down.

Table 11.1: Deluxe Chair Seat Troubleshooting (Sheet 2 of 2)

Possible Causes	Recommended Solution
Charging in progress	The seat height adjustment will not function while the charger is charging the battery. Wait until there is sufficient charge, and unplug the charger. The wired remote control has an LED indicator showing the charge status of the battery. If the left-most charging LED (green battery charge level indicator) is lit, then there should be enough power available to make at least one subject setup. NOTE: A fully charged battery should be able to complete approximately 120 full up-down cycles with a 80 kg load, or 70 cycles with a 150 kg load.
Seat lift motor has over-heated	The seat lift will automatically shut down if the motor overheats. The seat lift motor may overheat if it is run continuously for more than 2 minutes, if its use exceeds a 10% duty cycle (2 minutes total use within 20 minutes), or if the seat is loaded beyond the rated capacity while operating. Normal operation of the seat lift will resume once it detects that the motor has sufficiently cooled. However, the normal operating time will be reduced until the motor is fully cooled. It is recommended that the seat lift be given at least 30 minutes to cool down. If motion is resumed too soon, the robot lift will likely shut down again.

11.8 Poor Visibility in Virtual Reality Display

The Kinarm Labs Virtual Reality display is configured at the factory with a consistent brightness. However, if ambient light in the room is too high, the subject's ability to see in the Virtual Reality Display may be significantly reduced. Please refer to [6.5 Lighting](#) for recommendations to reduce the ambient lighting in the room.

12 Technical Specifications

Table 12.1: Technical Specifications of the Kinarm End-Point Lab
(Sheet 1 of 2)

Kinarm End-Point Lab	
Input voltage	Factory hardwired at one of the following: • 100 VAC, 50/60 Hz, or • 110-120 VAC, 50/60 Hz, or • 200 VAC, 50/60 Hz, or • 220-240 VAC, 50/60 Hz
Rated current	• 8 A continuous (15 A intermittent) for 100 or 110-120 VAC • 4 A continuous (7.5 A intermittent) for 200 or 220-240 VAC
Maximum subject mass (Standard Chair)	127 kg
Maximum subject mass (Deluxe Chair)	150 kg
Maximum drawer content mass (Power Rack)	5 kg
Kinarm End-Point Robot	
Maximum continuous end-point force (center of work-space)	19 N
Maximum intermittent (<5 s duration) end-point force (center of work-space)	58 N
Feedback resolution	2 μ m
Workspace (approximate elliptical workspace per robot)	76 x 44 cm
Effective inertia at end point	0.8/1.1 kg (minor/major axes)
End-Point Stiffness	44 N/mm
Force/Torque sensor maximum sensing range	80 N in XY plane
Force/Torque sensor overload range (i.e. before damage)	810 N in plane, 2400 N vertical, 19 Nm torque (about X and Y axes)
Force/Torque Sensor Resolution	0.02 N

Table 12.1: Technical Specifications of the Kinarm End-Point Lab
(Sheet 2 of 2)

Mass of robot (per robot)	35 kg
DAQ Options	
Robot Digital Inputs (delay between actual and recorded is 1.0 ms)	4 per robot
Robot Digital Outputs (delay between commanded and actual is 1.0 ms)	2 per robot
Optional Analog Inputs (16-bit)	16 or 32 channels
Optional Analog Outputs (16-bit)	2 or 4 channels
Optional Digital I/O	23 or 46 channels
Kinarm Gaze-Tracker	
Workspace	~ 55° x 40° ellipse
Resolution	~0.05°
Mean Accuracy with limited head movement	~0.5°
Rate	500 Hz
Environmental Conditions	
Operating temperature range	10 to 40 °C (excluding Kinarm Gaze-Tracker) 10 to 30 °C (Kinarm Gaze-Tracker)
Storage temperature range	-20 to 60 °C
Operating humidity range	20-80%
Storage humidity range	10-85%
Operating altitude (maximum)	2000 m